

URINARY SYSTEM AND FLUID BALANCE

ANATOMY AND PHYSIOLOGY (MC1)
COLLEGE OF NURSING
PRESIDENT RAMON MAGSAYSAY STATE UNIVERSITY

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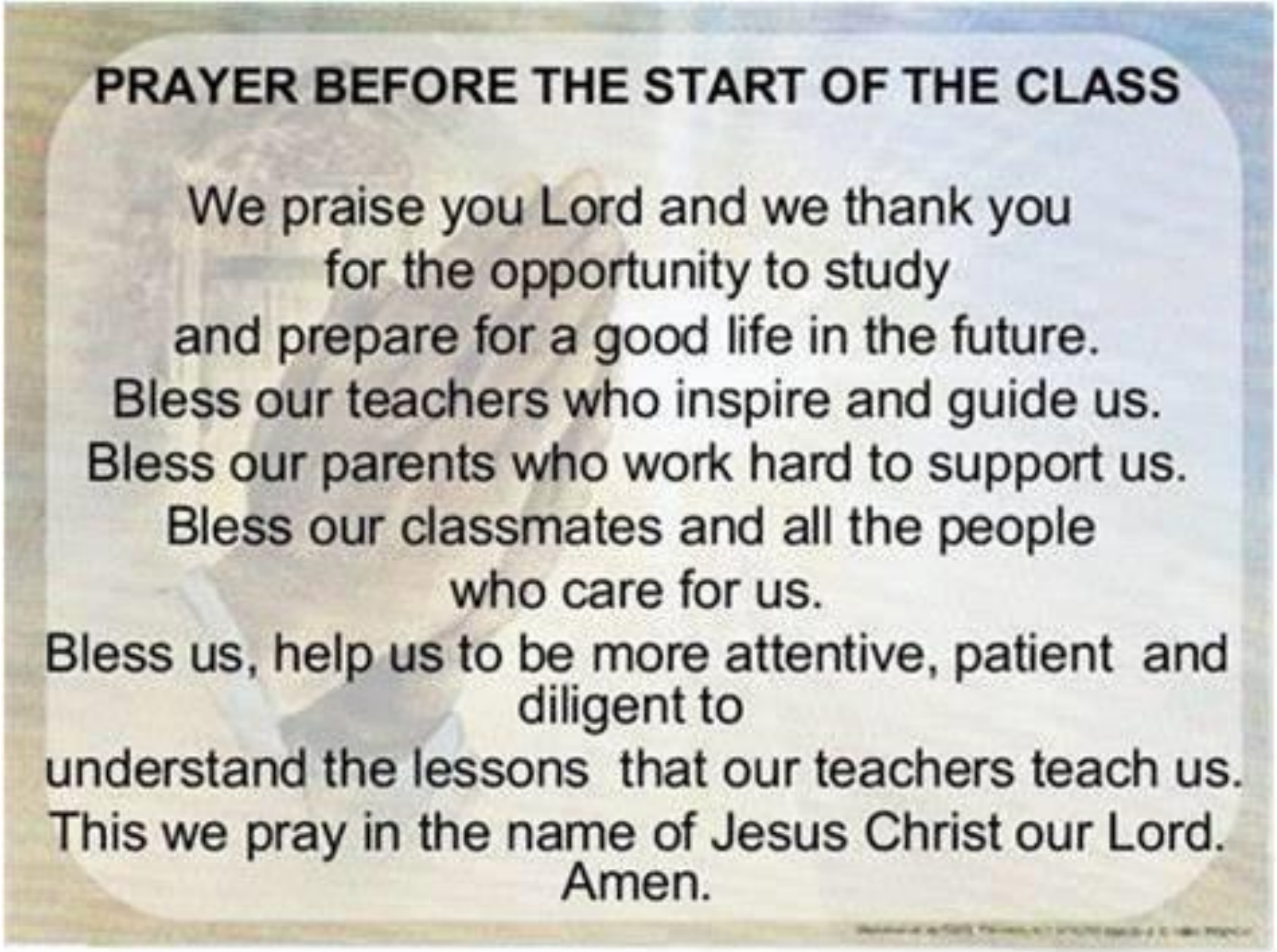
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ANATOMY AND PHYSIOLOGY

2020

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PRAYER BEFORE THE START OF THE CLASS
We praise you Lord and we thank you
for the opportunity to study
and prepare for a good life in the future.
Bless our teachers who inspire and guide us.
Bless our parents who work hard to support us.
Bless our classmates and all the people
who care for us.
Bless us, help us to be more attentive, patient and diligent to
understand the lessons that our teachers teach us.
This we pray in the name of Jesus Christ our Lord.
Amen.

OBJECTIVES

By the end of this lecture, students will be able to:

1. **Identify the major structures of the urinary system and describe their functions.**
2. **Explain the processes of urine formation: filtration, reabsorption, secretion, and excretion.**
3. **Describe how the kidneys contribute to fluid, electrolyte, and acid-base balance.**
4. **Discuss the hormonal regulation of fluid balance, particularly the roles of ADH and aldosterone.**
5. **Recognize the clinical significance of fluid imbalance and renal dysfunction.**
6. **Recognize the developmental aspects of the urinary system.**

FUNCTIONS OF THE URINARY SYSTEM

- ✓ EXCRETION
  - metabolic by products from intestines, skin, liver, lungs (nitrogenous wastes, toxins, drugs, excess ions)
- ✓ REGULATION OF BLOOD VOLUME AND PRESSURE
  - Kidneys control the ECF volume through production of diluted urine or small volume of concentrated urine
  - Control of renin
- ✓ REGULATION OF CONCENTRATION OF SOLUTES IN THE BLOOD
  - Regulates concentration of glucose, Na, Cl, K, Ca, HCO_3

FUNCTIONS OF THE URINARY SYSTEM

- ✓ REGULATION OF ECF pH
  - Secretion of H to regulate pH
- ✓ REGULATION OF RBC SYNTHESIS
  - Secretes erythropoietin, a hormone regulating the synthesis of RBC in bone marrow
- ✓ VIT D SYNTHESIS
  - Controls Ca blood level
  - Converts Vit D to its active form

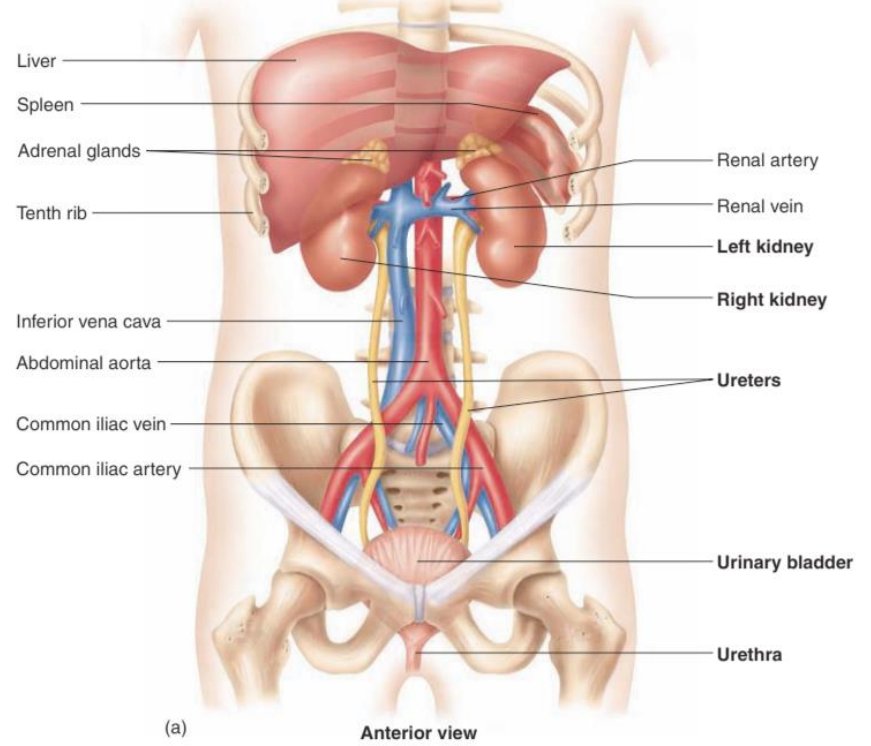
ORGANS AND STRUCTURES IN THE URINARY SYSTEM

- KIDNEYS
- URETERS
- URINARY BLADDER
- URETHRA

KIDNEYS

KIDNEYS: LOCATION

- The kidneys are in retroperitoneal position
- The kidneys are situated at the level of the T_{12} to L_3 vertebrae
- The right kidney is slightly lower than the left because of position of the liver



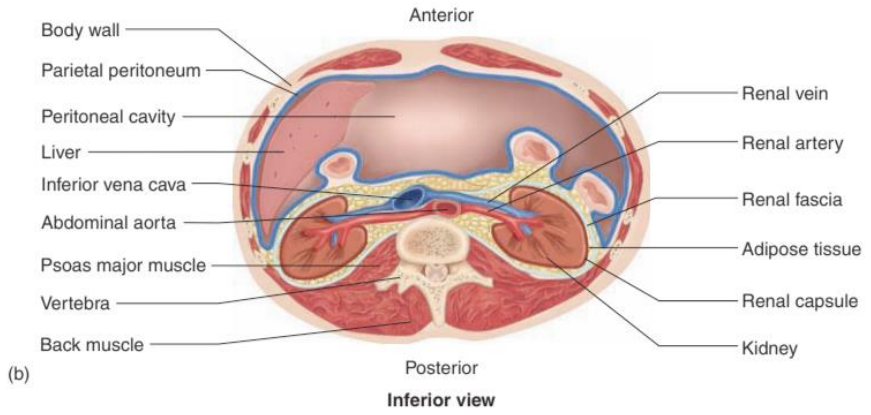


FIGURE 26.2 Anatomy of the Urinary System

(*a*) The kidneys are located in the abdominal cavity, with the right kidney just below the liver and the left kidney below the spleen. A ureter extends from each kidney to the urinary bladder within the pelvic cavity. An adrenal gland is located at the superior pole of each kidney. (*b*) The kidneys are located behind the parietal peritoneum, surrounded by adipose tissue. A connective tissue layer, the renal fascia, anchors the kidney to the abdominal wall. The renal arteries extend from the abdominal aorta to each kidney, and the renal veins extend from the kidneys to the inferior vena cava. AP|R

KIDNEY: STRUCTURE

- An adult kidney is about 12 cm (5 in) long and 6 cm (2.5 in) wide
- Bean shaped organs
- Size of a tightly clenched fist
- An adrenal gland sits atop each kidney

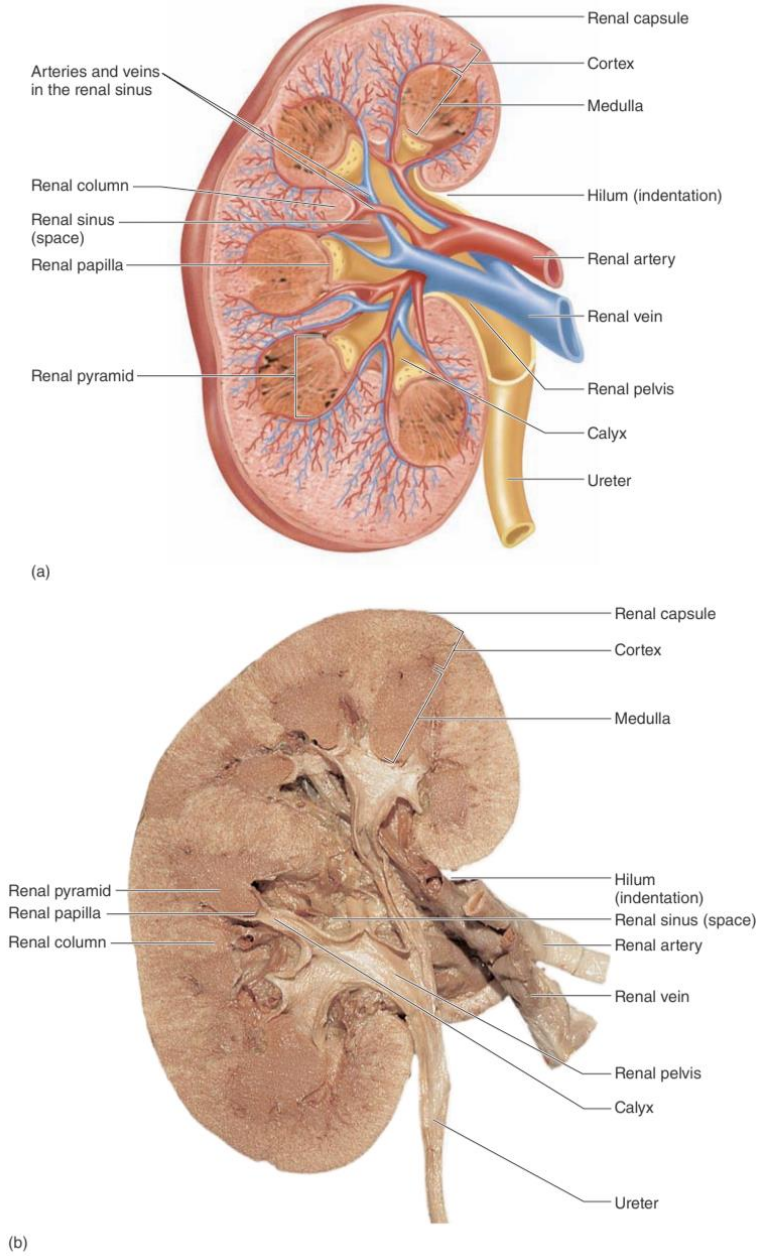
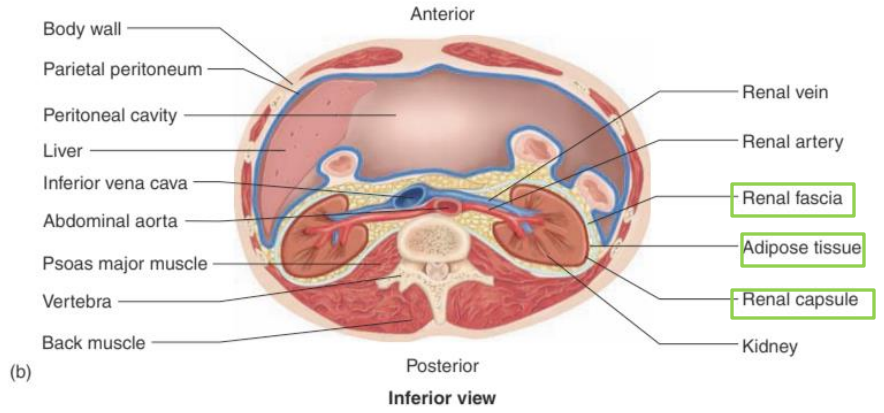
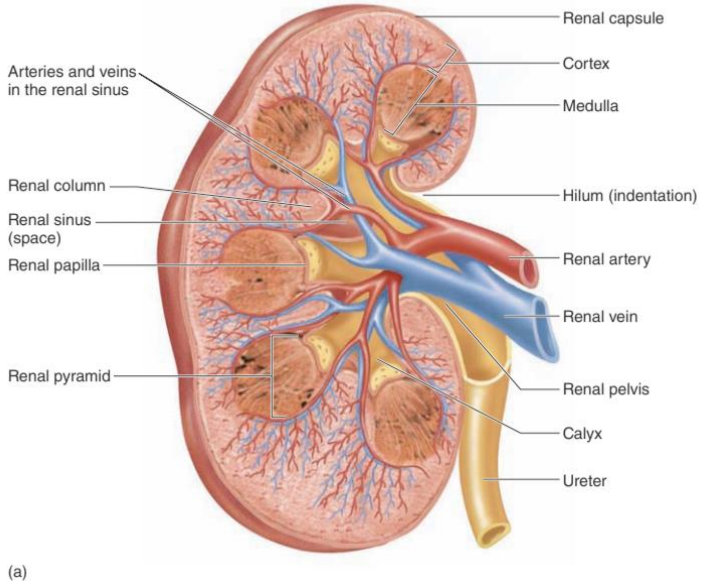


FIGURE 26.3 Frontal Section of the Kidney and Ureter
(*a*) A frontal kidney section shows that the cortex forms the outer part of the kidney, and the medulla forms the inner part. A central cavity called the renal sinus contains the renal pelvis. The renal columns of the kidney project from the cortex into the medulla and separate the pyramids. (*b*) Photograph of a longitudinal section of a human kidney and ureter. AP|R

KIDNEY: PARTS

- ✓ Three **PROTECTIVE LAYERS** enclose the kidney
  - Renal fascia
    - is the most superficial fat layer that anchors the kidney and adrenal gland to surrounding structures
    - Composed of anterior and posterior
  - Fibrous capsule
    - encloses each kidney
  - Perirenal fat capsule
    - surrounds the kidney and cushions against blows





KIDNEY: PARTS

- ✓ Renal hilum
  - A medial indentation where several structures enter or exit the kidney (ureters, renal blood vessels, and nerves)
- ✓ Renal Sinus
  - Cavity/ Opening containing the blood vessels and fat

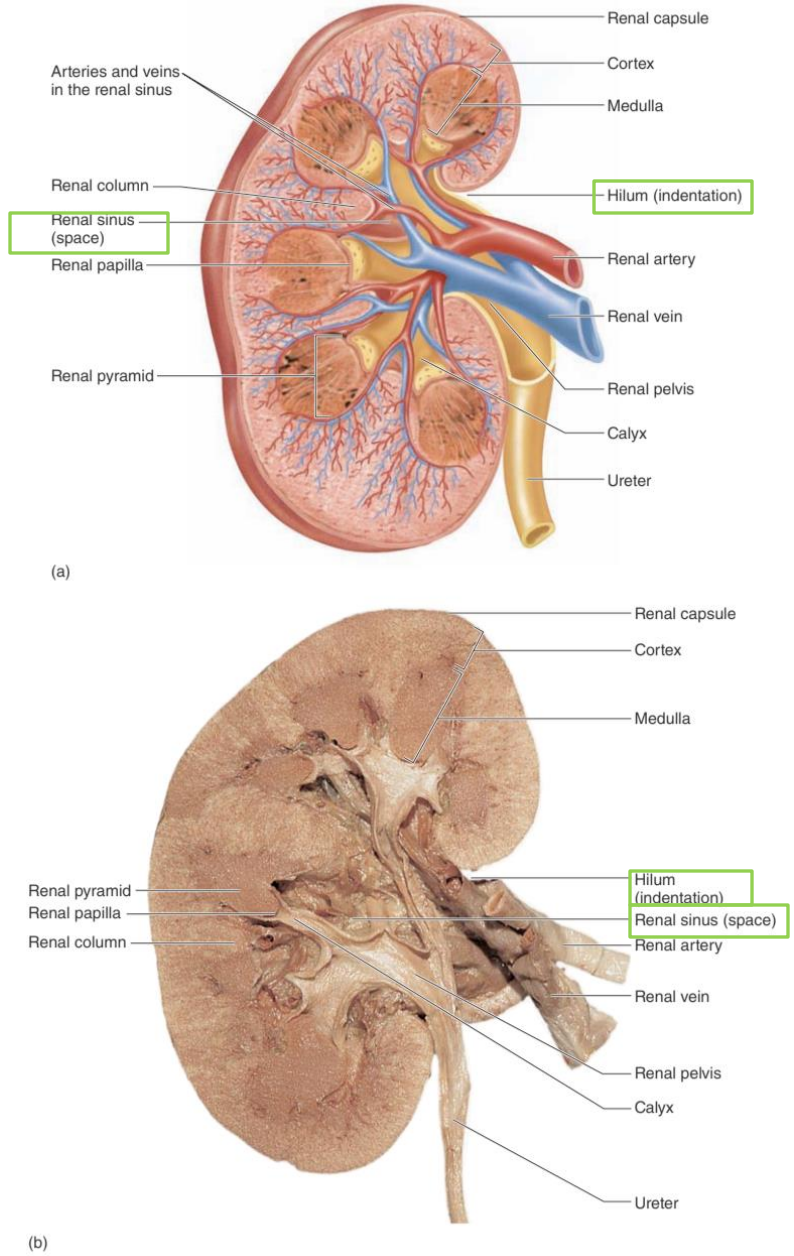


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KIDNEY: PARTS

- ✓ Three regions revealed in a longitudinal section
- Renal cortex—outer region
- Renal medulla—deeper region
  - Renal medullary pyramids—triangular regions of tissue in the medulla
  - Renal columns—extensions of cortex-like material that separate the pyramids
- Renal pelvis—medial region that is a flat, funnel-shaped tube
  - Composed of major and minor calyces
  - Calyces form cup-shaped "drains" that enclose the renal pyramids
  - Calyces collect urine and send it to the renal pelvis, on to the ureter, and to the urinary bladder for storage

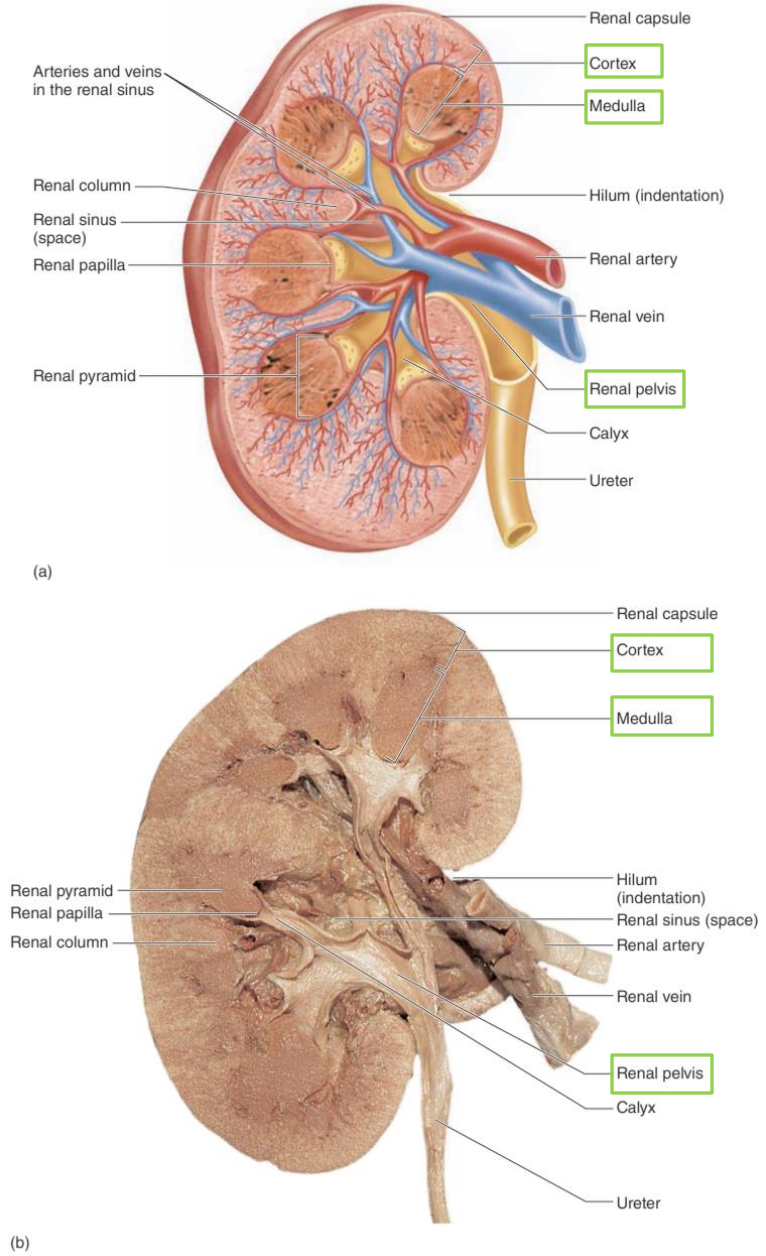


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KIDNEY: BLOOD SUPPLY

- ✓ **Arterial Blood Supply**
  - One-quarter of the total blood supply of the body passes through the kidneys each minute
  - Renal artery provides each kidney with arterial blood supply
  - Renal artery divides into segmental arteries → arcuate arteries → interlobar arteries → cortical radiate arteries

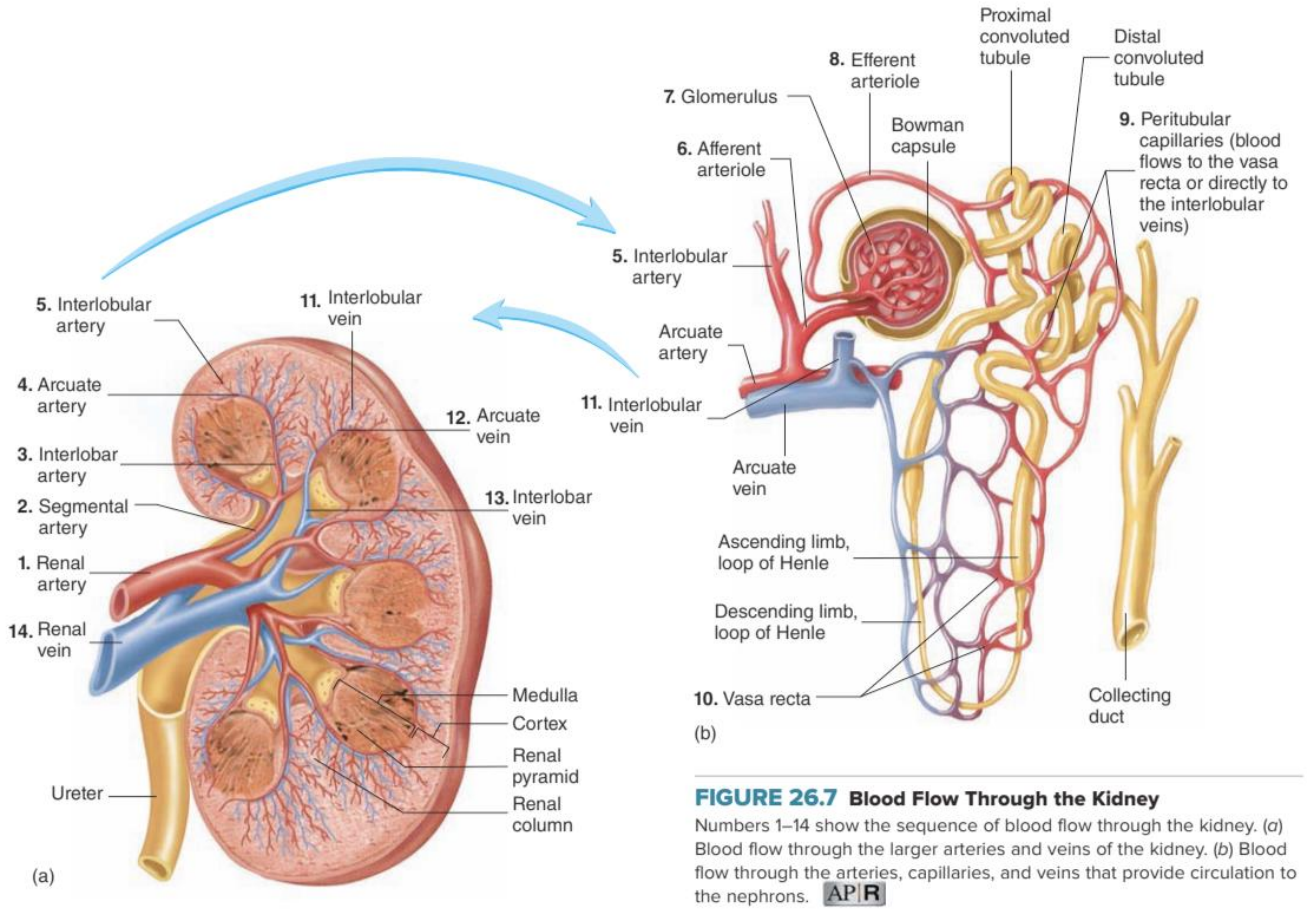


FIGURE 26.7 Blood Flow Through the Kidney
Numbers 1–14 show the sequence of blood flow through the kidney. (*a*) Blood flow through the larger arteries and veins of the kidney. (*b*) Blood flow through the arteries, capillaries, and veins that provide circulation to the nephrons. AP|R

KIDNEY: BLOOD SUPPLY

- ✓ **Venous blood flow**
  - Cortical radiate veins → arcuate veins → interlobar veins → renal vein
  - There are no segmental veins
  - Renal vein returns blood to the inferior vena cava

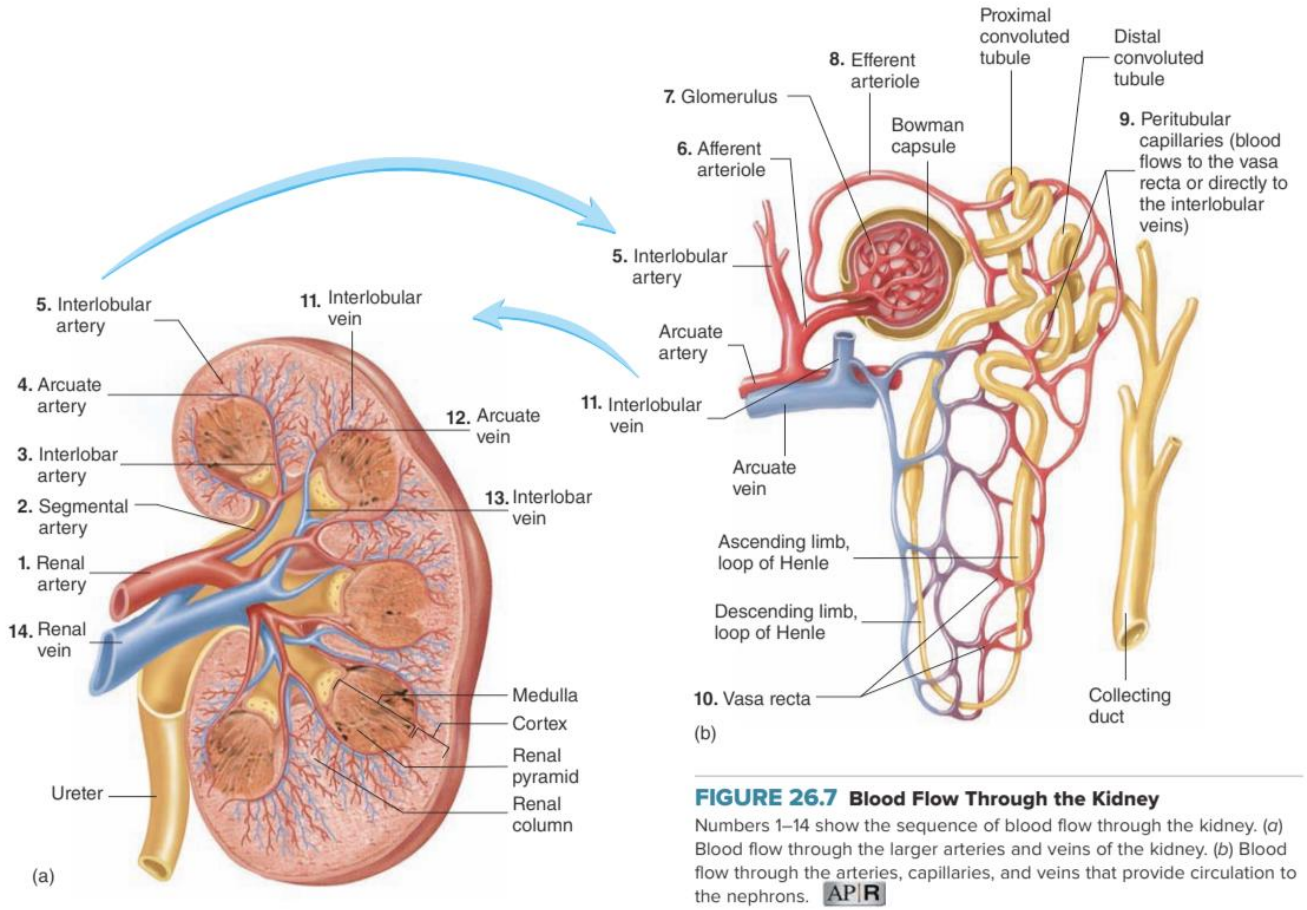
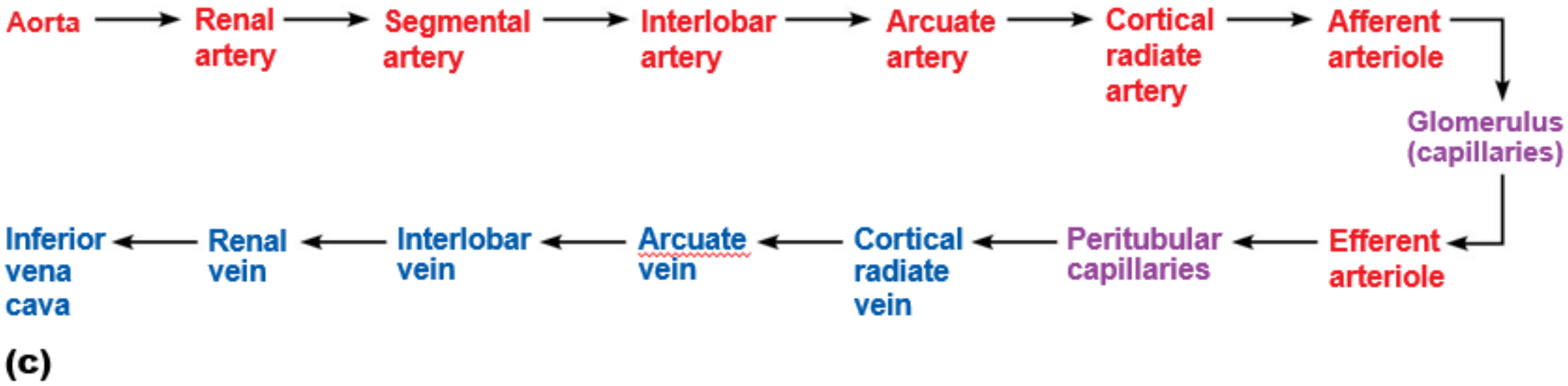


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KIDNEY: BLOOD SUPPLY



(c)

KIDNEY: NEPHRONS

- ✓ Structural and functional units of the kidneys
- ✓ Each kidney contains over a million nephrons
- ✓ Each nephron consists of two main structures
- ✓ Parts:
  - Renal corpuscle
    - consists of:
      - Glomerulus
      - Glomerular (Bowman's) capsule
  - Renal tubules
    - subdivisions:
      - Proximal convoluted tubule (PCT)
      - Nephron loop (loop of Henle)
      - Distal convoluted tubule (DCT)

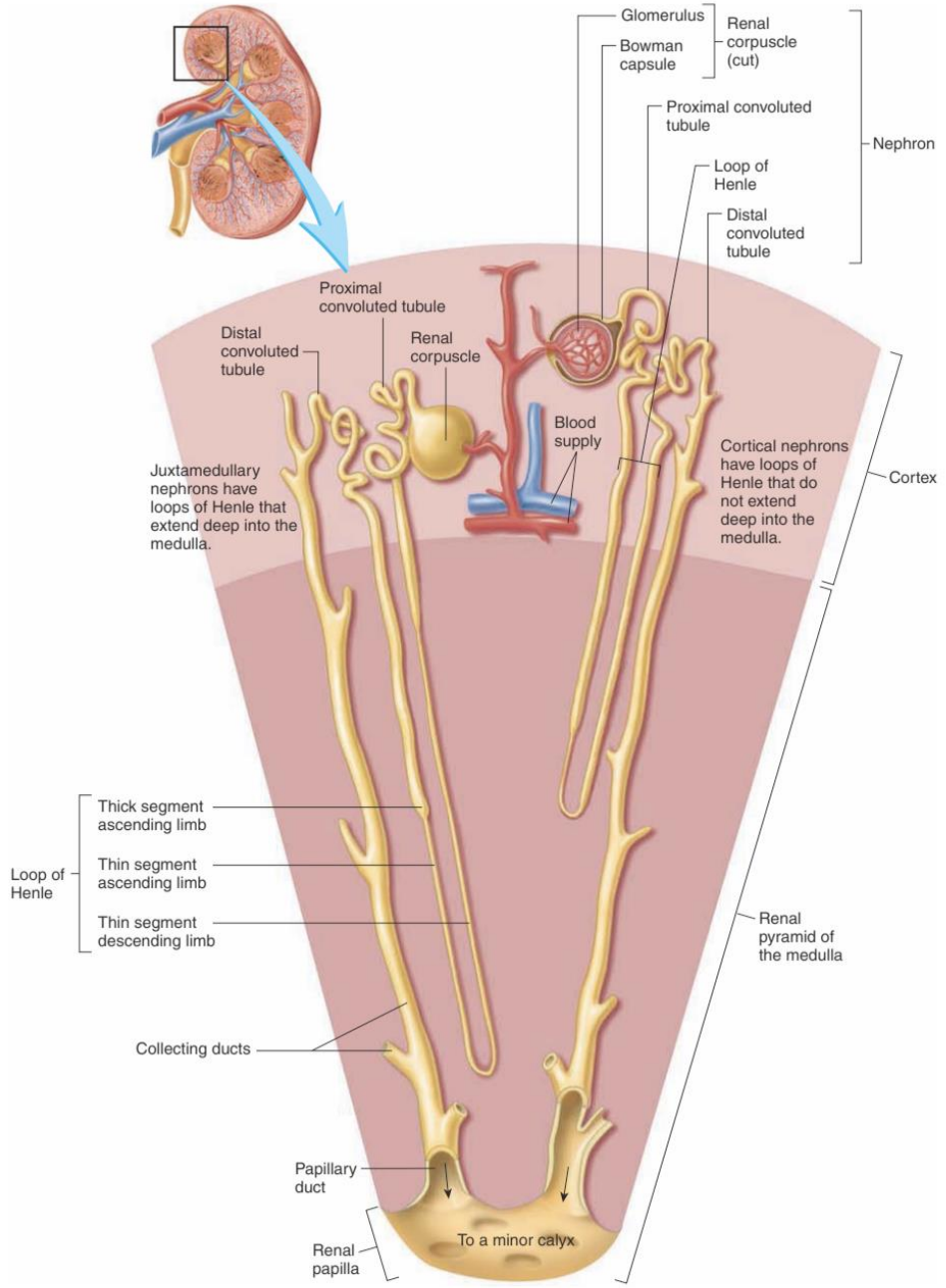


FIGURE 26.4 Functional Unit of the Kidney—the Nephron
A nephron consists of a renal corpuscle, proximal convoluted tubule, loop of Henle, and distal convoluted tubule. The distal convoluted tubule empties into a collecting duct. Juxtamedullary nephrons (those near the medulla of the kidney) have loops of Henle that extend deep into the medulla, whereas other nephrons do not. Collecting ducts undergo a transition to larger-diameter papillary ducts near the tip of the renal papilla. The papillary ducts empty into a minor calyx. AP|R

KIDNEY: NEPHRONS

- Renal corpuscle
  - consists of:
    - Glomerulus
      - a knot of capillaries made of *podocytes* which makes up the inner layer of the glomerular capsule
      - *Foot processes* cling to the glomerulus
      - *Filtration slits* create a porous membrane—ideal for filtration
    - Glomerular (Bowman's) capsule
      - a cup-shaped structure that surrounds the glomerulus
      - First part of the renal tubule

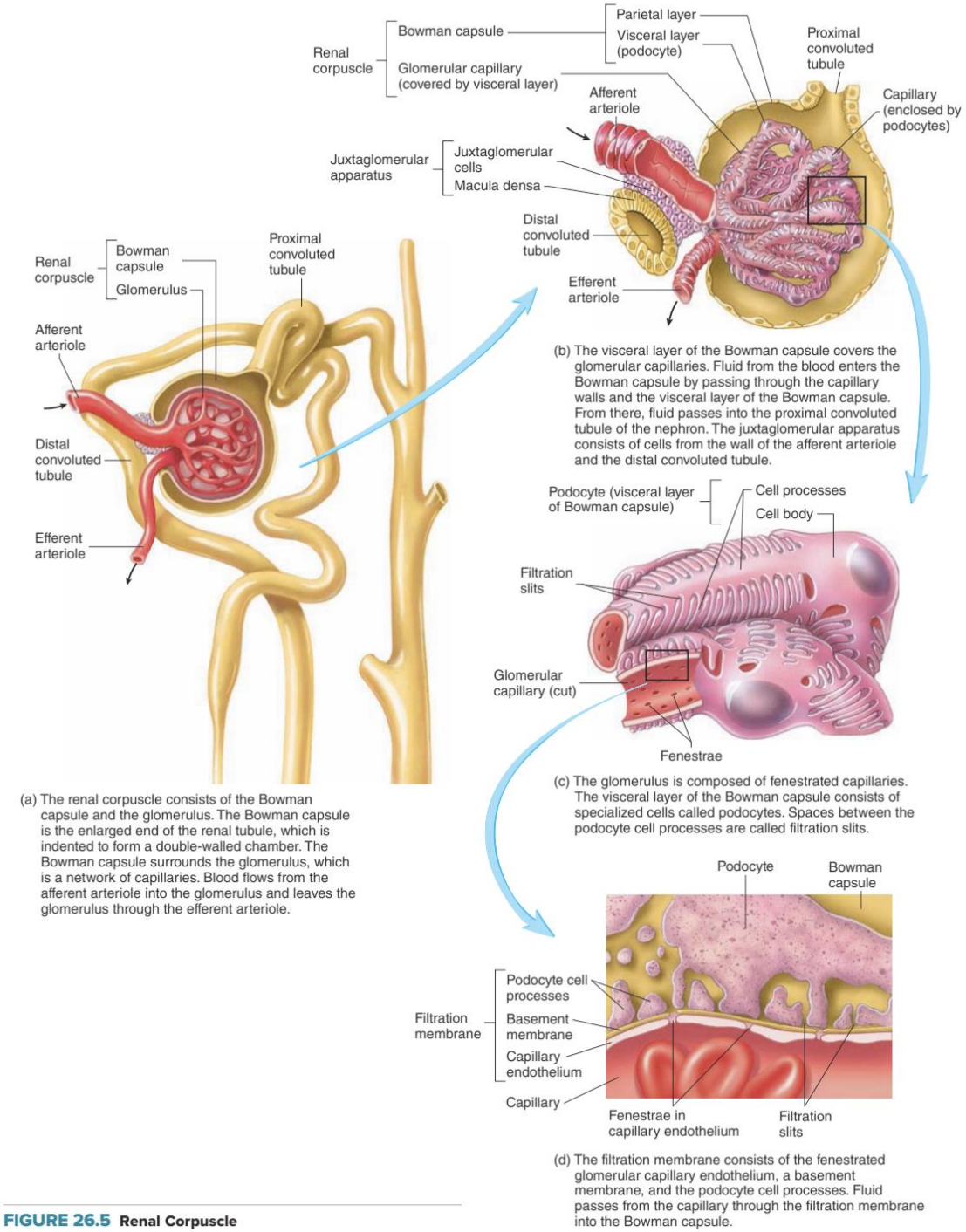


FIGURE 26.5 Renal Corpuscle

KIDNEY: NEPHRONS

- Renal tubules
  - Extends from glomerular capsule and ends when it empties into the collecting duct
  - From the glomerular (Bowman's) capsule, the subdivisions of the renal tubule are:
  - Proximal convoluted tubule (PCT)
  - Nephron loop (loop of Henle)
  - Distal convoluted tubule (DCT)

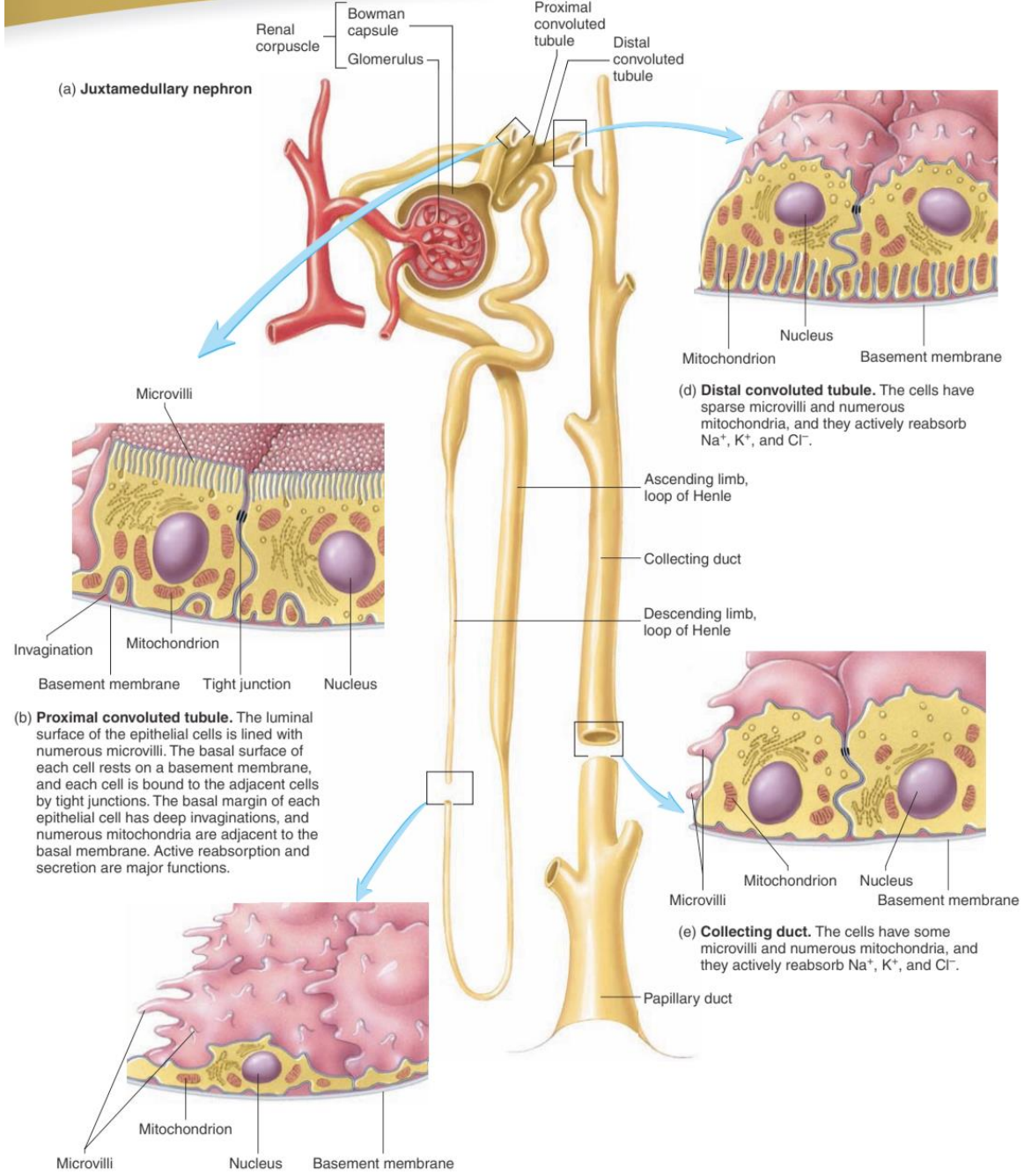
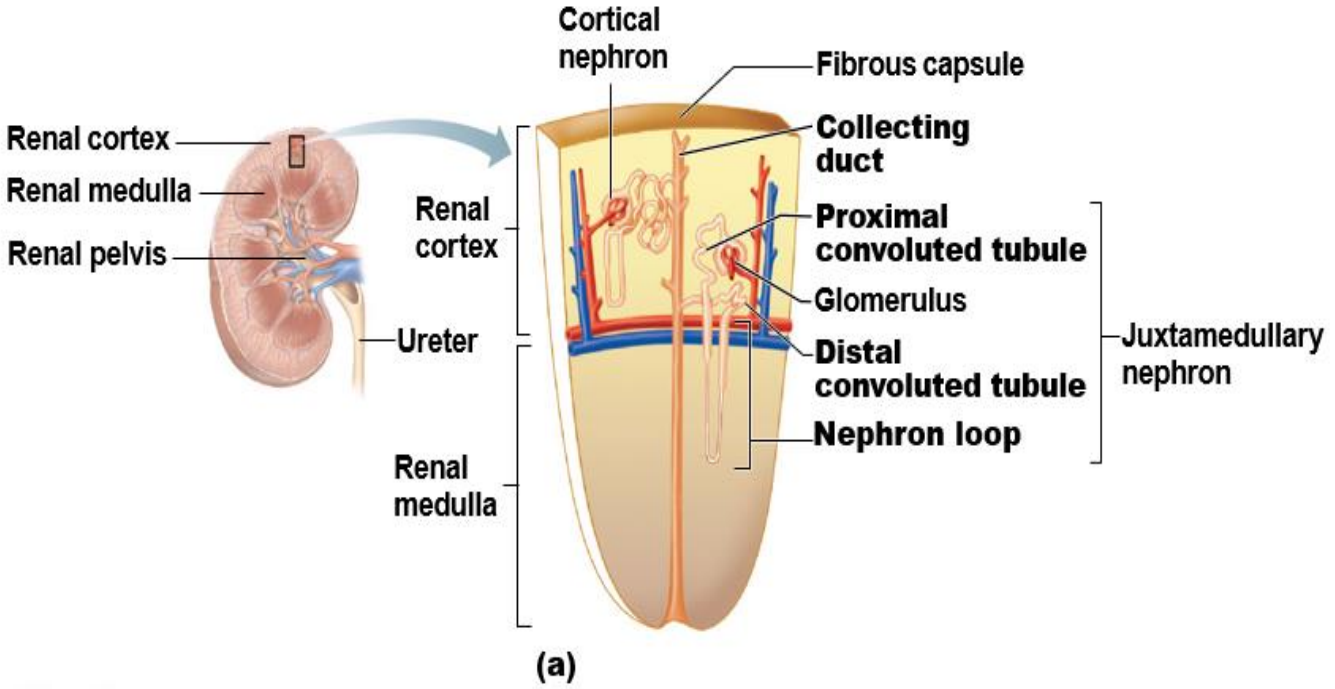


FIGURE 26.6 Histology of the Nephron

KIDNEY: NEPHRONS BASED ON LOCATION

- ✓ Cortical nephrons
  - Located entirely in the cortex
  - Include most nephrons
- ✓ Juxtamedullary nephrons
  - Found at the cortex-medulla junction
  - Nephron loop dips deep into the medulla



KIDNEY:
COLLECTING DUCTS

- Collecting ducts
  - collect urine from both types of nephrons, through the renal pyramids, to the calyces, and then to the renal pelvis

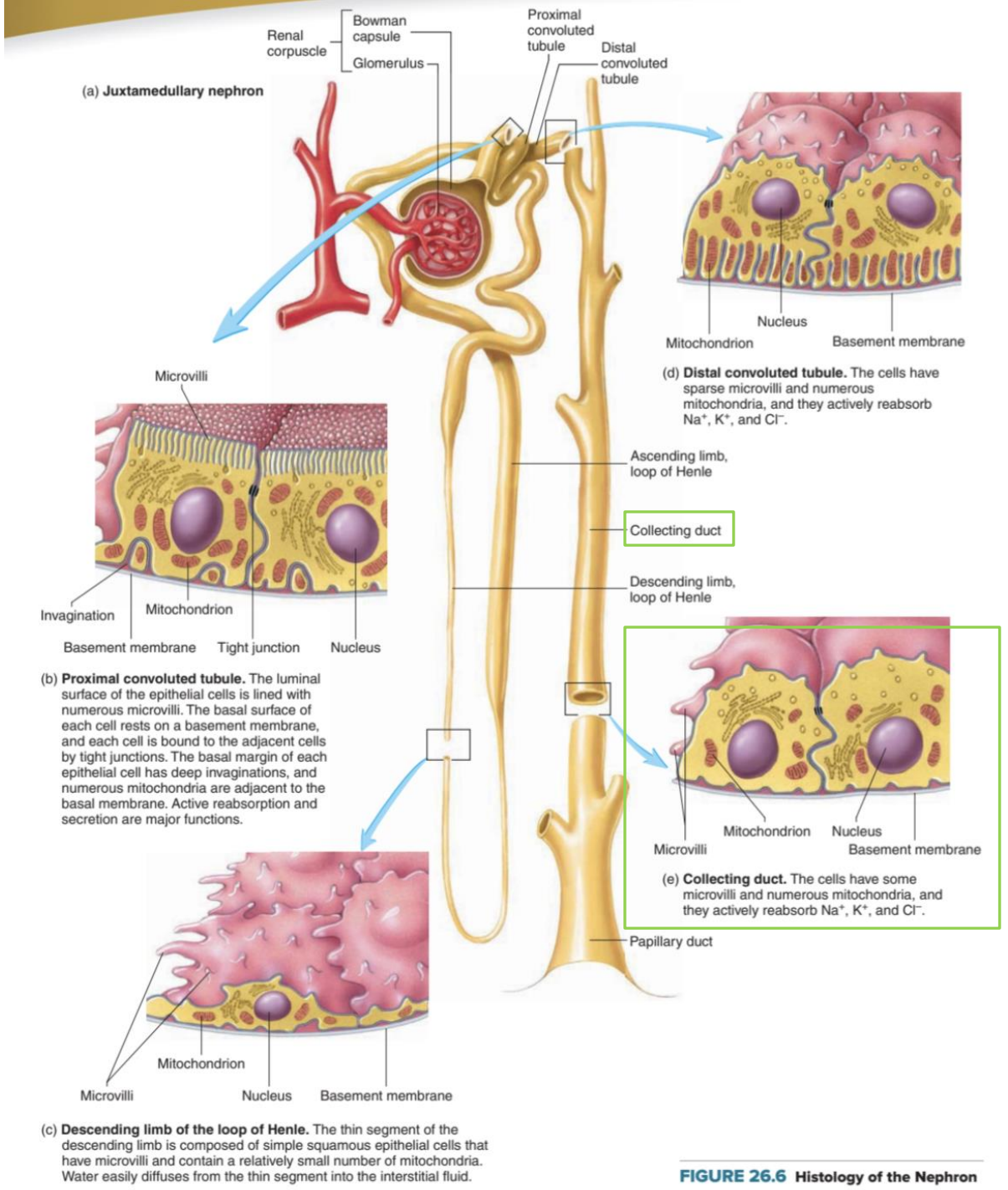


FIGURE 26.6 Histology of the Nephron

KIDNEY: TWO CAPILLARY BEDS ASSOCIATED WITH EACH NEPHRON

- ✓ Glomerulus
  - Fed and drained by arterioles
    - Afferent arteriole
    - Efferent arteriole
- ✓ Peritubular capillary beds

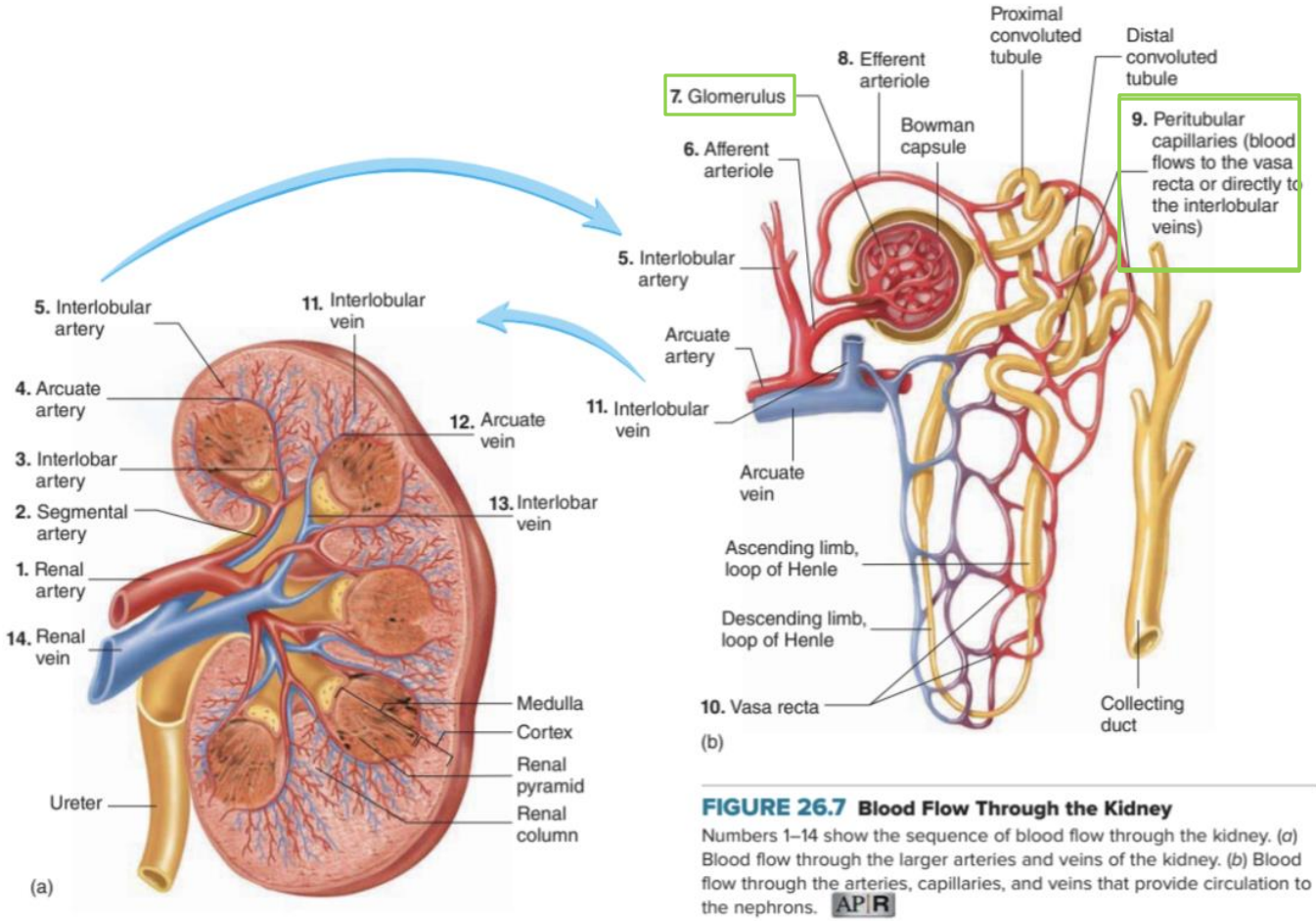


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KIDNEY: TWO CAPILLARY BEDS ASSOCIATED

- ✓ Glomerulus
  - Fed and drained by arterioles
    - Afferent arteriole—arises from a cortical radiate artery and feeds the glomerulus
    - Efferent arteriole—receives blood that has passed through the glomerulus
  - Specialized for <u>filtration</u>
  - High pressure forces fluid and solutes out of blood and into the glomerular capsule

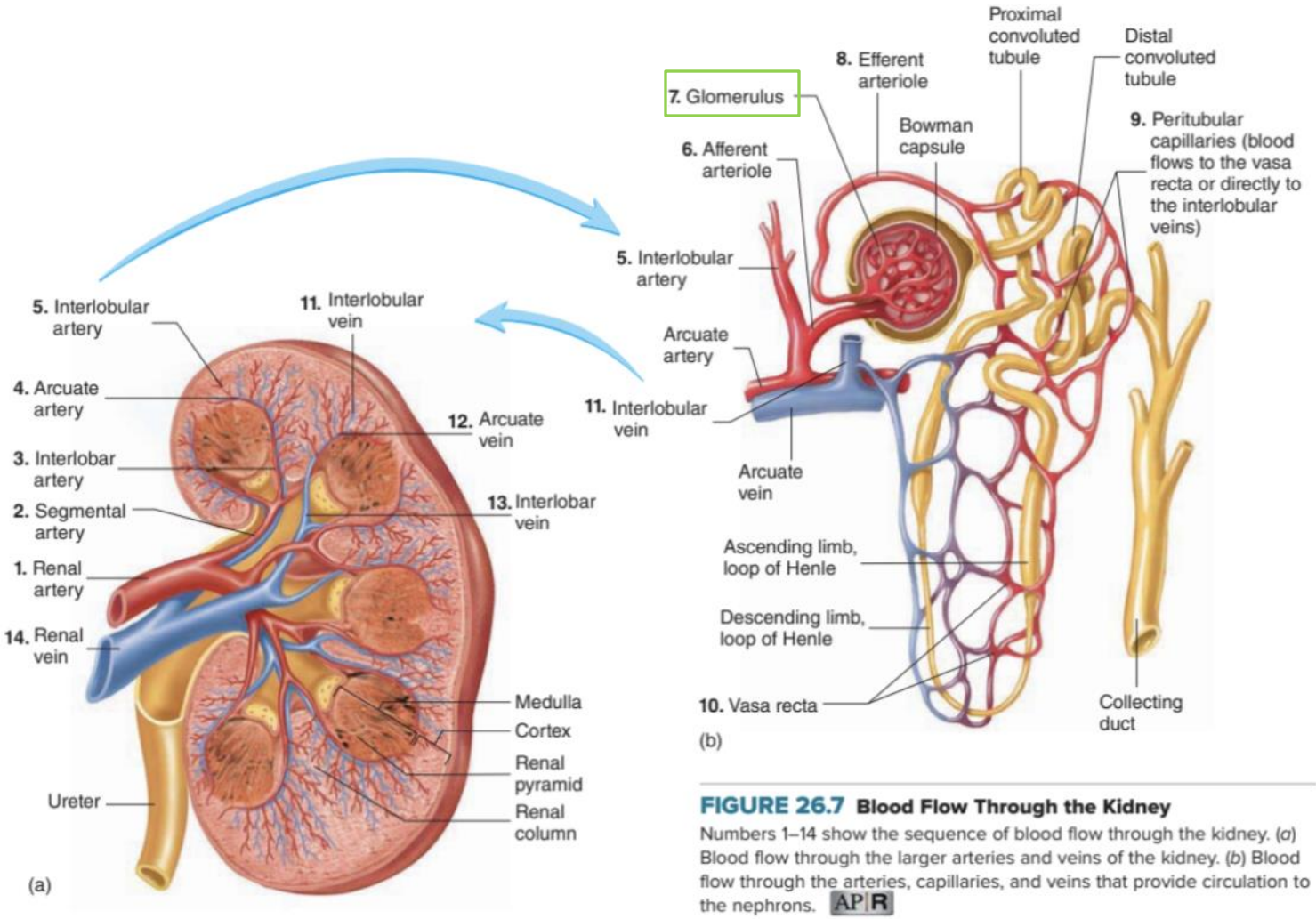


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KIDNEY: TWO CAPILLARY BEDS ASSOCIATED

- ✓ Peritubular capillary beds
  - Arise from the efferent arteriole of the glomerulus
  - Low-pressure, porous capillaries
  - Adapted for absorption instead of filtration
  - Cling close to the renal tubule to receive solutes and water from tubule cells
  - Drain into the interlobar veins

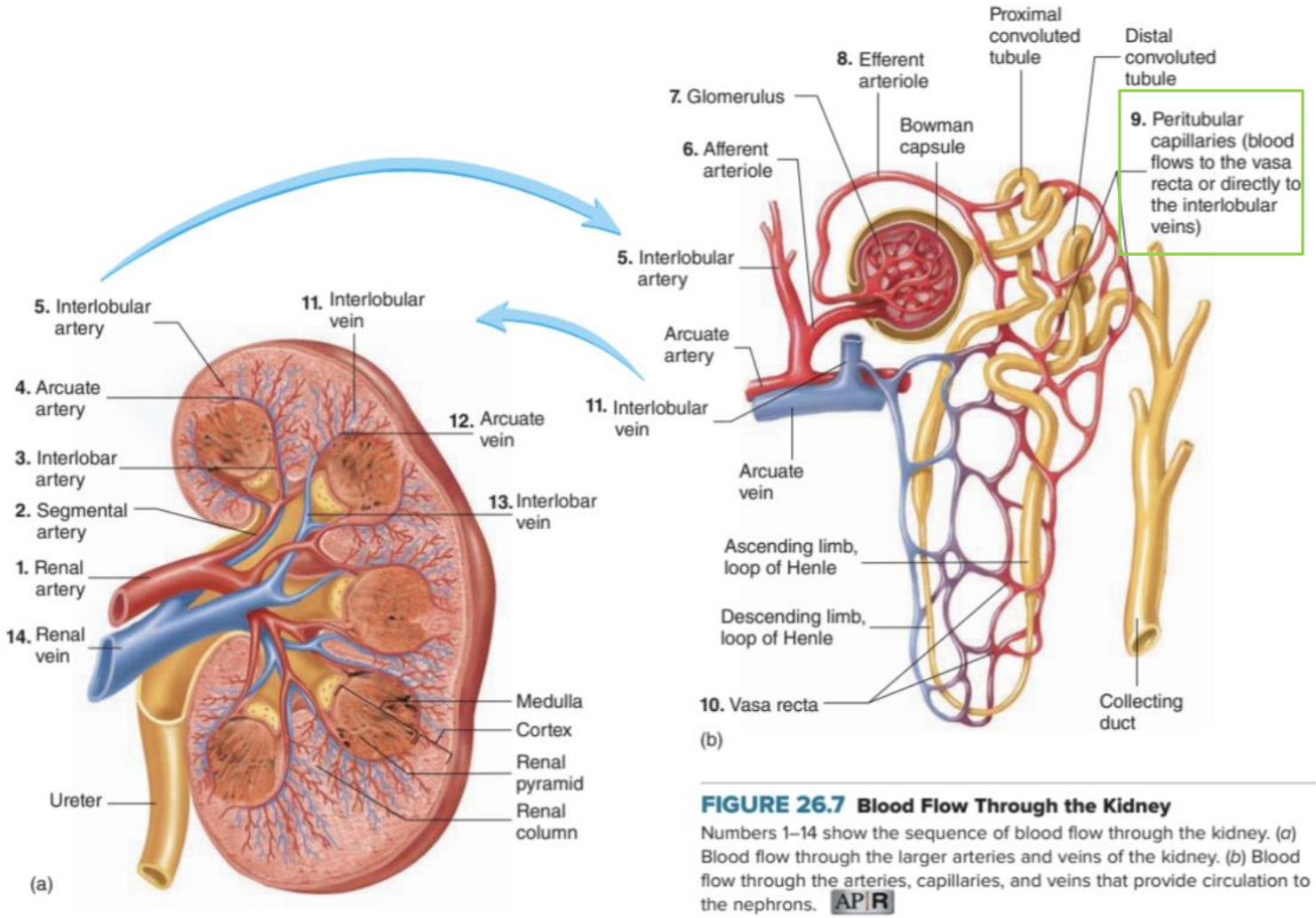


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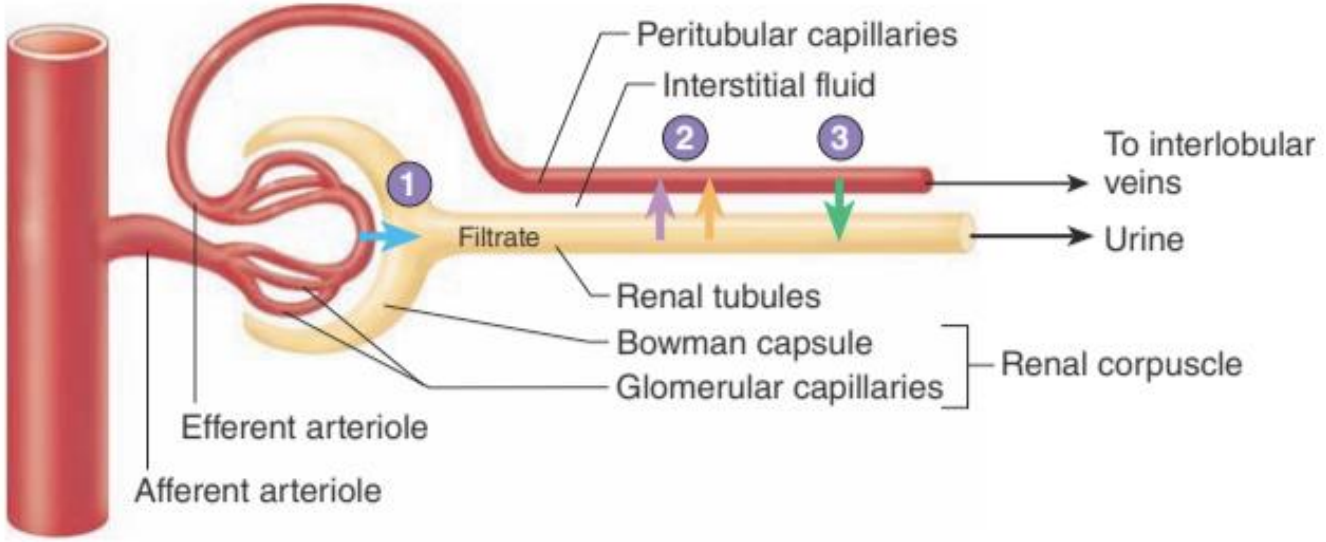
URINE FORMATION

- In 24 hours, about 1.0 to 1.8 liters of urine are produced
- Urine and filtrate are different
  - Filtrate contains everything that blood plasma does (except proteins)
  - Urine is what remains after the filtrate has lost most of its water, nutrients, and necessary ions through reabsorption
  - Urine contains nitrogenous wastes and substances that are not needed
    - Nitrogenous wastes
      - are poorly reabsorbed
      - tend to remain in the filtrate and are excreted from the body in the urine
        - Urea—end product of protein breakdown
        - Uric acid—results from nucleic acid metabolism
        - Creatinine—associated with creatine metabolism in muscles

KIDNEY: URINE FORMATION

Primary Function of the Kidney

- Regulates body fluid composition
- Filters blood: retains useful substances, removes waste



Three Major Steps in Urine Formation

1. **Filtration**
2. **Tubular Reabsorption**
3. **Tubular Secretion**

KIDNEY: URINE FORMATION

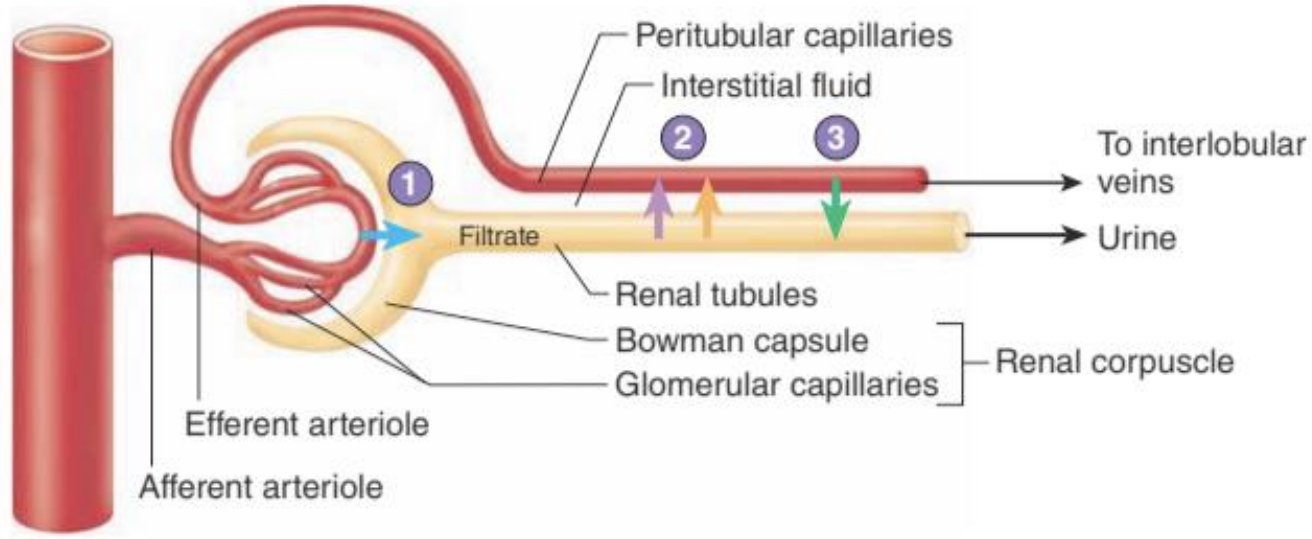
Urine production results from the following three processes:

1 Filtration — Filtration (*blue arrow*) is the movement of materials across the filtration membrane into the Bowman capsule to form filtrate.

2 Tubular reabsorption — Solutes are reabsorbed (*purple arrow*) across the wall of the renal tubule into the interstitial fluid by transport processes, such as active transport and cotransport.

Water is reabsorbed (*orange arrow*) across the wall of the renal tubule by osmosis. Water and solutes pass from the interstitial fluid into the peritubular capillaries.

3 Tubular secretion — Solutes are secreted (*green arrow*) across the wall of the renal tubule into the filtrate.



PROCESS FIGURE 26.8 Urine Production AP|R

KIDNEY: URINE FORMATION
1. FILTRATION

1. FILTRATION

- Nonspecific separation by **size & charge**
- Driven by **blood pressure,** not gravity
- Occurs in **renal corpuscle**
- Produces **filtrate** (precursor of urine)

KIDNEY: URINE FORMATION
1. FILTRATION

Renal Fraction

- Portion of the total cardiac output that flows through the kidneys
- Kidneys receive ~21% of cardiac output
- Indicates kidney's vital role in blood filtering

KIDNEY: URINE FORMATION
1. FILTRATION

Blood Flow and Filtrate Volume

- **Cardiac Output:** ~5600 mL/min
- **Renal Blood Flow:** 5600 × 0.21 = ~1176 mL/min
- **Plasma Flow Rate:** 1176 × 0.55 ≈ 650 mL/min
- **Filtration Fraction:** 650 × 0.19 ≈ 125 mL/min
- **GFR (Glomerular Filtration Rate):** ~125 mL/min or 180 L/day

KIDNEY: URINE FORMATION
1. FILTRATION

TABLE 26.2 Calculation of Renal Flow Rates

	Amount per Minute (mL)	Calculation
Renal Blood Flow	1176	Amount of blood flowing through the kidneys per minute; equals cardiac output (5600 mL blood/min) times the percentage (21%; renal fraction) of cardiac output that enters the kidneys 5600 mL blood/min × 0.21 = 1176 mL blood/min
Renal Plasma Flow	650	Amount of plasma flowing through the kidneys per minute; equals renal blood flow times percentage of the blood that is plasma. Because the hematocrit is the percentage of the blood that consists of formed elements, the percentage of the blood that is plasma is 100 minus the hematocrit. Assuming a hematocrit of 45, the percentage of the blood that is plasma is 55% (100 − 45). Renal plasma flow is therefore 55% of renal blood flow. 1176 mL blood/min × 0.55 ≈ 650 mL plasma/min
Glomerular Filtration Rate (GFR)	125	Amount of plasma (filtrate) that enters the Bowman capsule per minute; equals renal plasma flow times the percentage (19%; filtration fraction) of the plasma that enters the renal capsule 650 mL plasma/min × 0.19 ≈ 125 mL filtrate/min
Urine	1	Nonreabsorbed filtrate that leaves the kidneys per minute; equals glomerular filtration rate times the percentage (0.8%) of the filtrate that is not reabsorbed into the blood 125 mL filtrate/min × 0.008 = 1 mL urine/min Milliliters of urine per minute can be converted to liters of urine per day by multiplying by 1.44. 1 mL urine/min × 1.44 = 1.4 L/day

KIDNEY: URINE FORMATION
1. FILTRATION

Filtration Membrane Components

- Located in **renal corpuscle**
- Blocks most proteins and all blood cells
- Allows small molecules (e.g., glucose, ions, water)

Three key components:

1. **Fenestrae** in glomerular capillaries
2. **Basement membrane**
3. **Podocytes** of Bowman capsule

KIDNEY: URINE FORMATION
1. FILTRATION

Filtration Membrane: Size & Charge Selectivity

- Blocks molecules > **7 nm** or > **40,000 daltons**
- Allows:
  - Glucose (0.9 nm)
  - Methionine (0.7 nm)
- Repels negatively charged proteins (e.g., albumin)
- Filtrate contains **~0.03% protein**

KIDNEY: URINE FORMATION
1. FILTRATION

Filtration Pressure

Three pressures determine net filtration:

1. **Glomerular Capillary Pressure (GCP):** 50 mm Hg (outward)
2. **Capsular Hydrostatic Pressure (CHP):** 10 mm Hg (inward)
3. **Blood Colloid Osmotic Pressure (BCOP):** 30 mm Hg (inward)

📌 **Net Filtration Pressure (NFP):**
GCP – CHP – BCOP = **10 mm Hg**

KIDNEY: URINE FORMATION
1. FILTRATION

Clinical Insight – Glomerulonephritis

- kidney disorder where inflammation and damage occur in the glomeruli

Etiology:

- **Infections:** Post-infectious glomerulonephritis can occur after strep throat, skin infections, or other bacterial or viral infections
- **Autoimmune Diseases:** Conditions like systemic lupus erythematosus (SLE), vasculitis
- **Other Causes:** Drugs, toxins, certain cancers, and genetic factors

Pathophysiology:

- **Inflammation of glomeruli** leading to damage to the filtering structures
- **Increased Permeability of** Filtration membrane allowing protein blood cells to leak into the urine
- **Fluid Retention due to the kidney's** struggle to regulate fluid balance , leading to swelling (edema) in the face, legs, lungs and other areas
- **High Blood Pressure**

Signs and Symptoms:

- **Swelling (Edema)**
- **High Blood Pressure (Hypertension)**
- **Blood in the Urine (Hematuria):** Urine may appear pink, brown, or cola-colored.
- **Protein in the Urine (Proteinuria):** Urine may appear foamy or bubbly.
- **Decreased Urine Output**
- **Other Symptoms:** Fatigue, nausea, loss of appetite, joint pain, and weakness can also occur

KIDNEY: URINE FORMATION
1. FILTRATION

Regulation of Glomerular Filtration Rate (GFR)

- GFR remains stable between **90–180 mm Hg** MAP
- Maintained by **autoregulation**
- **Drops significantly** during hemorrhage or severe dehydration

KIDNEY: URINE FORMATION
1. FILTRATION

Regulation of Glomerular Filtration Rate (GFR): Autoregulation

TWO MECHANISMS OF AUTOREGULATION:

1. Myogenic Mechanism

- **Afferent arteriole smooth muscle** detects stretch
- ↑ BP → Stretch → **Vasoconstriction**
- ↓ BP → Relaxation → **Vasodilation**

2. Tubuloglomerular Feedback

- Based on filtrate flow past **macula densa**
- ↑ Filtrate flow → Macula densa signals afferent arteriole → **Vasoconstriction**
- ↓ Glomerular capillary pressure → ↓ GFR

KIDNEY: URINE FORMATION
1. FILTRATION

Regulation of Glomerular Filtration Rate (GFR): Sympathetic Stimulation

- Mild stimulation → **Little effect**
- Intense stimulation (e.g., **shock, exercise**) →
  - **Vasoconstriction** of renal vessels
  - ↓ Renal blood flow & ↓ GFR

KIDNEY: URINE FORMATION
1. FILTRATION

Glomerular Filtration Rate in Shock & Stress

- Intense vasoconstriction → **Minimal urine formation**
- Long-term ↓ blood flow → **Kidney damage**
- Short-term ↓ blood flow → **Protective (homeostatic)**

KIDNEY: URINE FORMATION
1. FILTRATION

Summary

- GFR stable due to **autoregulation**
- Myogenic & tubuloglomerular feedback are key
- Sympathetic activity reduces GFR under stress
- Prolonged reduction in flow → **renal damage**

KIDNEY: URINE FORMATION
2. TUBULAR REABSORPTION

2. TUBULAR REABSORPTION

- Return of water and solutes from filtrate to blood
- Approximately 99% of filtrate is reabsorbed
- Essential for maintaining fluid and electrolyte balance
- Prevents loss of vital nutrients

KIDNEY: URINE FORMATION
2. TUBULAR REABSORPTION

Mechanisms of Reabsorption

- Passive Transport: Diffusion, Osmosis
- Active Transport: Requires ATP
- Secondary Active Transport: Coupled with Na+ movement
- Facilitated Diffusion: Carrier proteins assist

KIDNEY: URINE FORMATION
2. TUBULAR REABSORPTION

Mechanisms of Reabsorption: Proximal Convoluted Tubule (PCT)

- Reabsorbs ~65% of filtrate volume
- Reabsorbed substances: Na^+, K^+, Cl^-, HCO_3^-, glucose, amino acids
- Water follows solutes via osmosis
- Utilizes Na^+/K^+ ATPase pumps

KIDNEY: URINE FORMATION
2. TUBULAR REABSORPTION

Mechanisms of Reabsorption: Loop of Henle

Descending Limb:

- Permeable to water
- Water exits, filtrate becomes concentrated

Ascending Limb:

- Impermeable to water
- Na^+, K^+, Cl^- actively reabsorbed
- Filtrate becomes dilute

KIDNEY: URINE FORMATION
2. TUBULAR REABSORPTION

Mechanisms of Reabsorption: Distal Convoluted Tubule (DCT) and Collecting Duct

Reabsorbs Na+, Cl−, and Ca2+

Under hormonal control: Aldosterone and ADH

ADH increases water permeability

Fine-tunes electrolyte and fluid balance

KIDNEY: URINE FORMATION
2. TUBULAR REABSORPTION

Mechanisms of Reabsorption: Urea and Other Solutes

- Urea: ~50% passively reabsorbed
- Creatinine, uric acid: minimally reabsorbed
- Concentration increases as water is reabsorbed
- Eliminated to prevent toxicity

KIDNEY: URINE FORMATION
2. TUBULAR REABSORPTION

Summary

- Tubular reabsorption is vital for conserving essential substances
- Different nephron segments specialize in reabsorbing specific substances
- Hormonal regulation ensures adaptability to the body's needs
- Efficient reabsorption maintains homeostasis and prevents dehydration

KIDNEY: URINE FORMATION
3. TUBULAR SECRETION

3. TUBULAR SECRETION

- Movement of substances **from blood → filtrate**
- Removes:
  - **Metabolic by-products**
  - **Drugs & toxins**
- **Ions (e.g., H^+, K^+)**
- Occurs in **PCT & DCT**
- Involves **active and passive transport**

KIDNEY: URINE FORMATION
3. TUBULAR SECRETION

3. TUBULAR SECRETION: Substances Secreted

Actively Secreted:

- H^+
- K^+
- Penicillin
- PAH (para-aminohippuric acid)

Passively Secreted:

- Ammonia (NH_3): Peritubular capillaries → tubule cells → lumen

KIDNEY: URINE FORMATION
3. TUBULAR SECRETION

3. TUBULAR SECRETION: Ammonia Secretion

- **Ammonia (NH_3):** by-product of amino acid metabolism
- Produced by **renal tubule epithelial cells**
- Diffuses into the **tubular lumen**
- Helps buffer **excess H^+**

KIDNEY: URINE FORMATION
3. TUBULAR SECRETION

3. TUBULAR SECRETION: Antiport Mechanism – H^+ and Na^+

- **Antiport (exchange):**
  - Na^+ enters tubule cell
  - H^+ secreted into lumen
- **Driven by Na^+ gradient**
- Maintains **pH balance**

KIDNEY: URINE FORMATION
3. TUBULAR SECRETION

3. TUBULAR SECRETION: H^+ Generation and Secretion

- $CO_2 + H_2O \rightarrow H^+ + HCO_3^-$ (via carbonic anhydrase)
- H^+: Secreted into tubule lumen
- HCO_3^-: Reabsorbed into peritubular capillaries
- **Helps regulate blood pH**

KIDNEY: URINE FORMATION
3. TUBULAR SECRETION

3. TUBULAR SECRETION: Functional Significance

- Eliminates:
- Waste products not filtered
  - Excess ions (H^+, K^+)
  - Drugs and toxins
- **Maintains pH homeostasis**
- Fine-tunes filtrate composition

KIDNEY: URINE FORMATION
3. TUBULAR SECRETION

Summary

- Tubular secretion = blood → tubule lumen
- Removes wastes, drugs, and excess ions
- Uses active and passive mechanisms
- Crucial for pH regulation and detoxification

KIDNEY: REGULATION OF URINE CONCENTRATION AND VOLUME

URINE REGULATION

Urine Variability: Urine can be dilute or concentrated, produced in large or small volumes.

Homeostasis: The kidneys adjust urine concentration and volume to maintain extracellular fluid balance.

Key Processes:

- **Obligatory Reabsorption:** Occurs in the proximal convoluted tubules and descending limbs of the loops of Henle; remains relatively constant.
- **Regulated Reabsorption:** Occurs in the distal convoluted tubules and collecting ducts; varies based on body conditions.

KIDNEY: REGULATION OF URINE CONCENTRATION AND VOLUME

HORMONAL MECHANISMS IN URINE REGULATION

- **Renin-Angiotensin-Aldosterone System (RAAS):**
  - Activated by decreased blood pressure or sodium levels.
  - Leads to vasoconstriction and sodium retention.
- **Antidiuretic Hormone (ADH):**
  - Released in response to increased blood osmolality or decreased blood pressure.
  - Promotes water reabsorption in the kidneys.

KIDNEY: REGULATION OF URINE CONCENTRATION AND VOLUME

HORMONAL MECHANISMS IN URINE REGULATION:

Renin-Angiotensin-Aldosterone Mechanism

Renin Release:

- Triggered by low afferent arteriole pressure or low sodium at the macula densa.

Angiotensin II Formation:

- Renin converts angiotensinogen to angiotensin I.
- Angiotensin-converting enzyme (ACE) converts angiotensin I to angiotensin II.

Effects of Angiotensin II:

- Vasoconstriction increases blood pressure.
- Stimulates aldosterone secretion, thirst, and ADH release.

KIDNEY: REGULATION OF URINE CONCENTRATION AND VOLUME

HORMONAL MECHANISMS IN URINE REGULATION:

Aldosterone Function

- **Source:** Secreted by the adrenal cortex.
- **Action:**
  - Increases sodium reabsorption in distal convoluted tubules and collecting ducts.
  - Promotes potassium and hydrogen ion secretion.
- **Regulation:**
  - Stimulated by angiotensin II and high potassium levels.
  - Inhibited by high sodium levels.

KIDNEY: REGULATION OF URINE CONCENTRATION AND VOLUME

HORMONAL MECHANISMS IN URINE REGULATION:

Antidiuretic Hormone (ADH) Mechanism

- **Source:** Produced by the hypothalamus; released from the posterior pituitary.
- **Stimuli for Release:**
  - Increased blood osmolality.
  - Decreased blood pressure detected by baroreceptors.
- **Action:**
  - Increases water permeability in distal convoluted tubules and collecting ducts.
  - Promotes water reabsorption, concentrating urine.

KIDNEY: REGULATION OF URINE CONCENTRATION AND VOLUME

PRODUCTION OF CONCENTRATED URINE

- **ADH Presence:**
  - Inserts aquaporin-2 channels into the apical membranes of collecting duct cells.
  - Facilitates water reabsorption into the hyperosmotic medullary interstitium.
- **Result:**
  - Urine becomes concentrated (up to 1200 mOsm/kg).
  - Only about 1% of the original filtrate volume is excreted.

KIDNEY: REGULATION OF URINE CONCENTRATION AND VOLUME

PRODUCTION OF DILUTE URINE

- **ADH Absence:**
  - Collecting ducts remain impermeable to water.
  - Less water is reabsorbed.
- **Result:**
  - Urine remains dilute.
  - Larger volume of urine is excreted (up to 20–30 L/day in extreme cases).

KIDNEY: REGULATION OF URINE CONCENTRATION AND VOLUME

ATRIAL NATRIURETIC HORMONE (ANH)

- **Source:** Secreted by right atrial cardiac muscle cells in response to increased blood volume.
- **Actions:**
  - Inhibits sodium reabsorption in the kidneys.
  - Suppresses ADH secretion.
  - Increases urine volume, reducing blood volume and pressure.

KIDNEY: REGULATION OF URINE CONCENTRATION AND VOLUME

Summary

- **Kidney Regulation:**
  - Balances urine concentration and volume through hormonal and neural mechanisms.
- **Hormonal Controls:**
  - RAAS and ADH conserve water and sodium.
  - ANH promotes excretion of water and sodium.
- **Homeostasis:**
  - These systems work together to maintain fluid and electrolyte balance.

KIDNEY: URINE CHARACTERISTICS

Normal Characteristics of Urine:

- **Color:** Clear and pale to deep yellow
  - Due to *urochrome*, a pigment from hemoglobin breakdown
- **Clarity:** Transparent when freshly voided
- **Odor:** Slightly aromatic; develops ammonia scent over time
- **pH:** Slightly acidic (~6), but can range 4.5–8
- **Specific Gravity:** 1.001–1.035
  - Indicates urine concentration
- **Sterility:** Sterile at formation (in the kidneys)

Note: Pale straw color indicates dilute urine; deeper yellow indicates concentration.

KIDNEY: SOLUTES IN NORMAL URINE

- **Electrolytes:**
  - **Sodium (Na^+)** and **Potassium (K^+)** – regulate fluid balance and nerve/muscle function
- **Nitrogenous Wastes:**
  - **Urea** – from amino acid breakdown
  - **Uric Acid** – from nucleic acid metabolism
  - **Creatinine** – from muscle metabolism
- **Other Solutes:**
  - **Ammonia (NH_3)** – formed from protein metabolism
  - **Bicarbonate (HCO_3^-)** – helps regulate acid-base balance

These solutes reflect kidney filtration and overall metabolic activity.

KIDNEY: SOLUTES *NOT* NORMALLY FOUND IN URINE

- **Glucose** – Indicates possible **diabetes mellitus**
- **Blood Proteins (e.g., albumin)** – May suggest **kidney damage**
- **Red Blood Cells (RBCs)** – Sign of **trauma, infection, or stones**
- **Hemoglobin** – May result from **hemolytic anemia**
- **White Blood Cells (WBCs/pus)** – Indicates **urinary tract infection (UTI)**
- **Bile Pigments** – Associated with **liver disease or bile duct blockage**

Presence of these substances often signals underlying pathology.

KIDNEY: REGULATION OF URINE CONCENTRATION AND VOLUME CASE STUDY

Case Scenario: Amanda's Marathon Experience

- Amanda ran her first marathon in 35°C heat with low humidity.
- She drank very little water during the race.
- After 4½ hours: dizziness, disorientation, tachycardia, pallor.
- Diagnosed with severe dehydration; received IV fluids.
- Did not urinate until 12 hours later.

Question:
Why did Amanda produce very little urine?

KIDNEY: REGULATION OF URINE CONCENTRATION AND VOLUME CASE STUDY

Case Scenario:

Why did Amanda produce very little urine?

ANSWER:

Amanda's case illustrates a classic **physiological response** to **severe dehydration.**

1. Fluid Loss Without Replacement:

- During prolonged physical exertion in high temperatures, Amanda lost a significant amount of fluid through **sweat and respiration.**
- However, she **did not adequately hydrate,** so her blood volume and plasma water content **dropped**—a state called **hypovolemia.**

KIDNEY: REGULATION OF URINE CONCENTRATION AND VOLUME CASE STUDY

Case Scenario:

Why did Amanda produce very little urine?

ANSWER:

Amanda's case illustrates a classic **physiological response** to **severe dehydration.**

2. ADH (Antidiuretic Hormone) Response:

- As Amanda's blood became more concentrated (**increased osmolality**), **osmoreceptors in the hypothalamus** were triggered.
- This led to **increased ADH release** from the posterior pituitary.
- ADH increased **water reabsorption** in the distal convoluted tubules and collecting ducts by inserting **aquaporin-2 channels.**
- Result: **Minimal water lost in urine,** leading to **concentrated urine and reduced volume.**

KIDNEY: REGULATION OF URINE CONCENTRATION AND VOLUME CASE STUDY

Case Scenario:

Why did Amanda produce very little urine?

ANSWER:

Amanda's case illustrates a classic **physiological response** to **severe dehydration.**

3. Renin-Angiotensin-Aldosterone System (RAAS):

- Her kidneys detected **low blood pressure and sodium** via the **juxtaglomerular apparatus**.
- Renin was released, initiating the formation of **angiotensin II**.
- Angiotensin II caused **vasoconstriction** and stimulated the **release of aldosterone** from the adrenal cortex.
- Aldosterone promoted **sodium reabsorption**, which pulled more water back into the blood.
- This helped raise blood pressure and volume, but further **reduced urine output**.

KIDNEY: REGULATION OF URINE CONCENTRATION AND VOLUME CASE STUDY

Case Scenario:

Why did Amanda produce very little urine?

ANSWER:

Amanda's case illustrates a classic **physiological response** to **severe dehydration.**

4. Reduced Urine Production:

- Due to **maximum water reabsorption** and reduced glomerular filtration rate (GFR), Amanda's kidneys produced **very little urine.**
- She did not urinate for nearly **12 hours**—a protective mechanism to preserve fluid and maintain circulation.

KIDNEY: REGULATION OF URINE CONCENTRATION AND VOLUME CASE STUDY

Case Scenario:

Why did Amanda produce very little urine?

ANSWER:

Amanda's case illustrates a classic **physiological response** to **severe dehydration.**

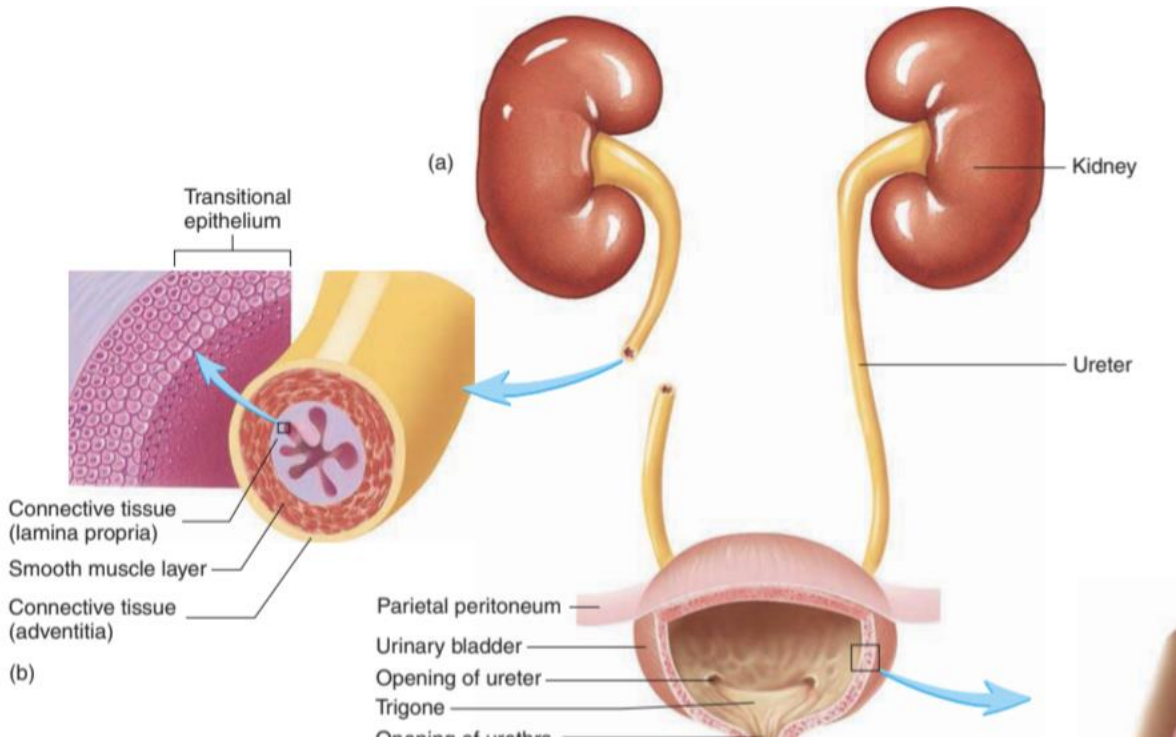
Clinical Outcome:

- IV fluids helped restore **blood volume and electrolyte balance.**
- Once hydration and perfusion were reestablished, normal **urine production resumed.**

URETERS

URETERS

- Slender muscular tubes: **25–30 cm (10–12 inches)**
- Connect **renal pelvis** to **urinary bladder**
- Enter the **posterior aspect** of the bladder
- Travel **behind the peritoneum**
- **Peristalsis** assists gravity in moving urine



URINARY BLADDER

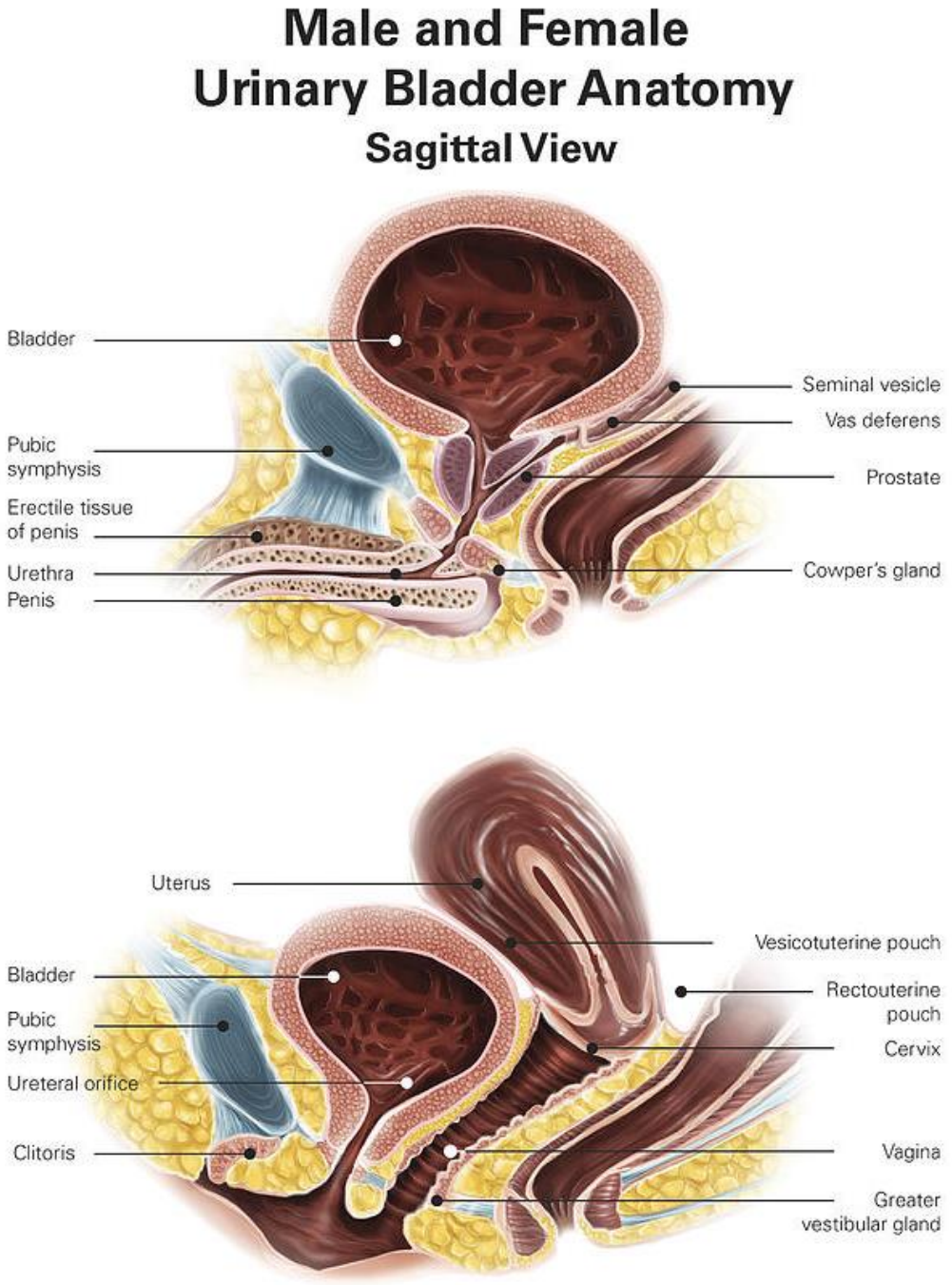
URINARY BLADDER

- Smooth, collapsible, muscular sac behind the **pubic symphysis**
- **Temporarily stores urine**

Parts:

- **Trigone**: Triangular area formed by:
  - 2 **ureteral openings**
  - 1 **internal urethral orifice**

- In males, **prostate** surrounds the bladder neck
- **Capacity:**
  - ~500 ml = moderately full
  - Max ~1000 ml



URETHRA

URETHRA

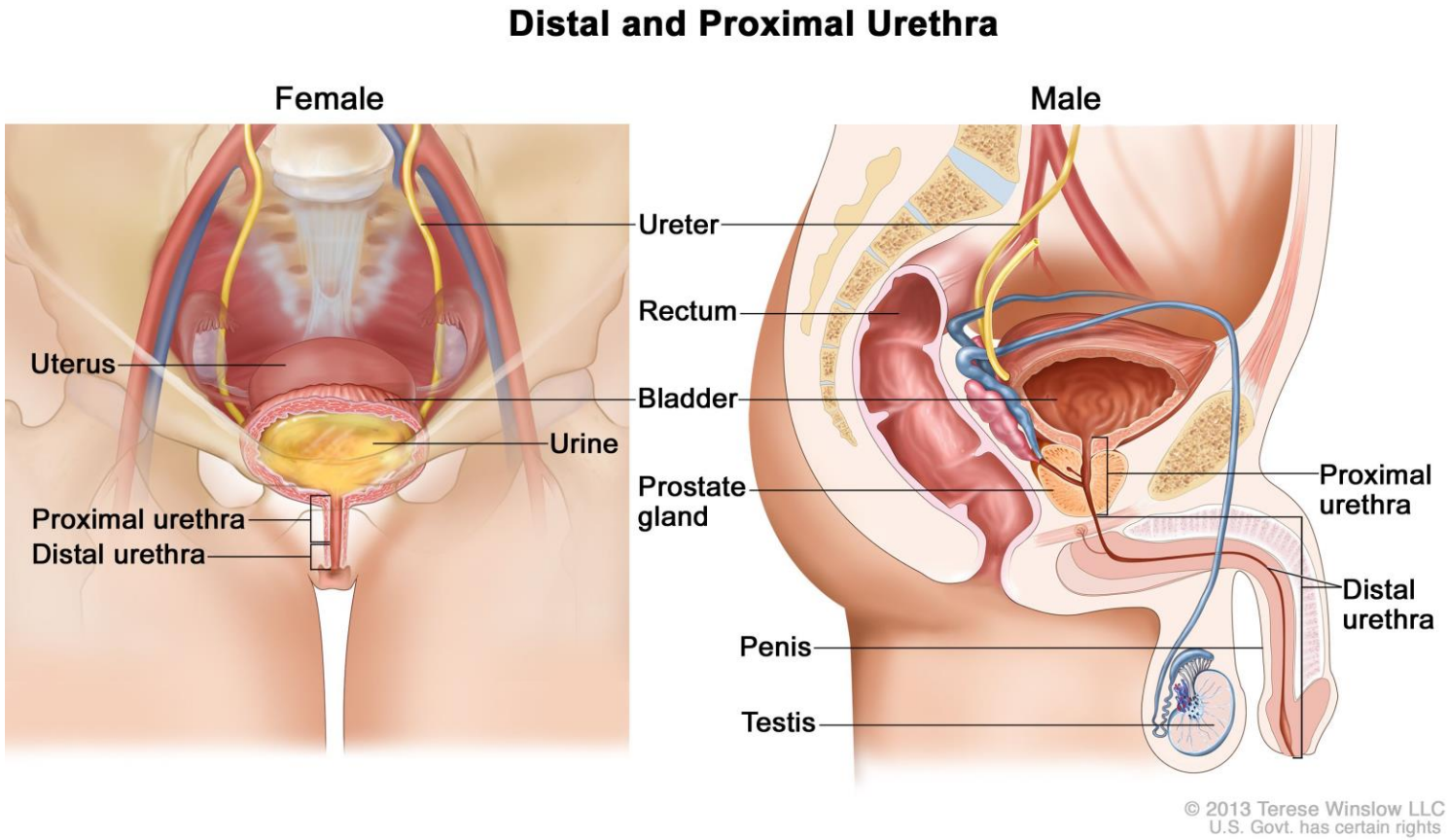
- Thin-walled tube that expels urine from bladder to outside via **peristalsis**

Function:
- **Females:** Carries urine only
- **Males:** Carries both **urine and sperm**

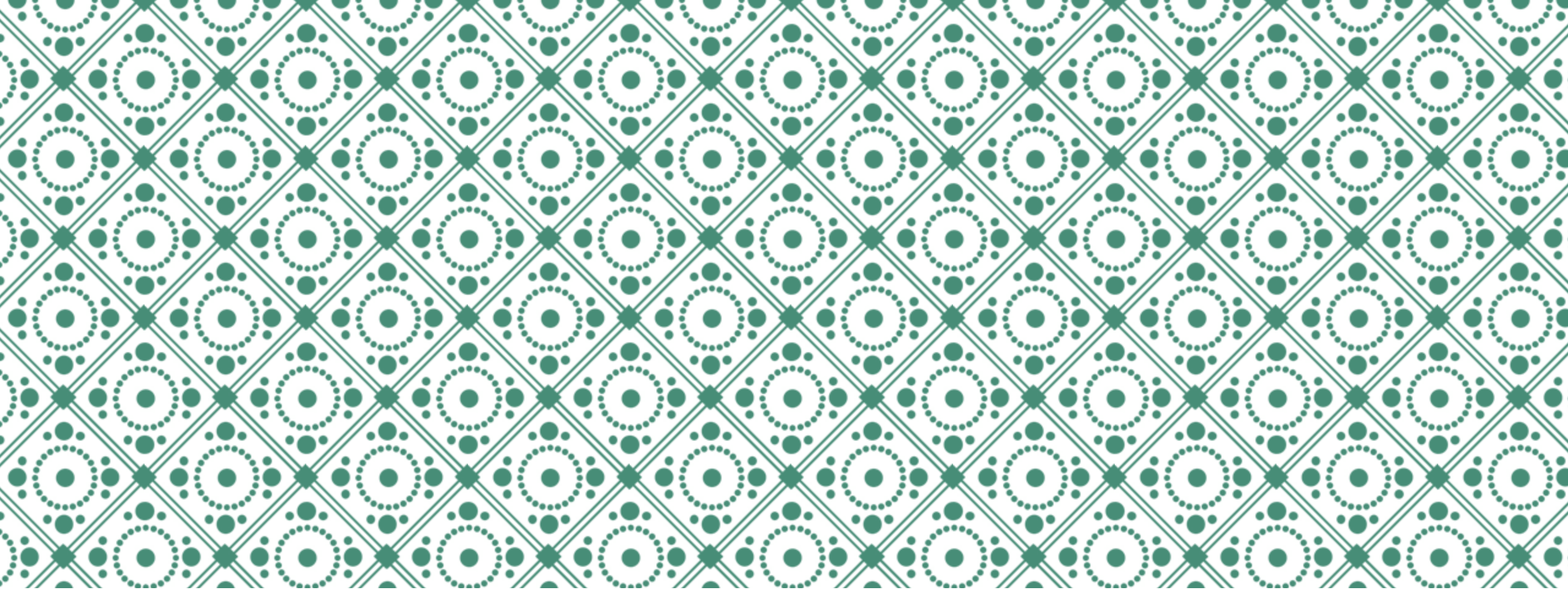
Length:
- **Females:** 3–4 cm (1.5 in)
- **Males:** 20 cm (8 in)

Location:
- **Females:** Anterior to vaginal opening
- **Males:** Passes through:
  - **Prostatic urethra**
  - **Membranous urethra**
  - **Spongy urethra**

Sphincters controlling urination:
- **Internal urethral sphincter:** Involuntary, smooth muscle
- **External urethral sphincter:** Voluntary, skeletal muscle



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MICTURITION

MICTURITION

Micturition: The Urination Process

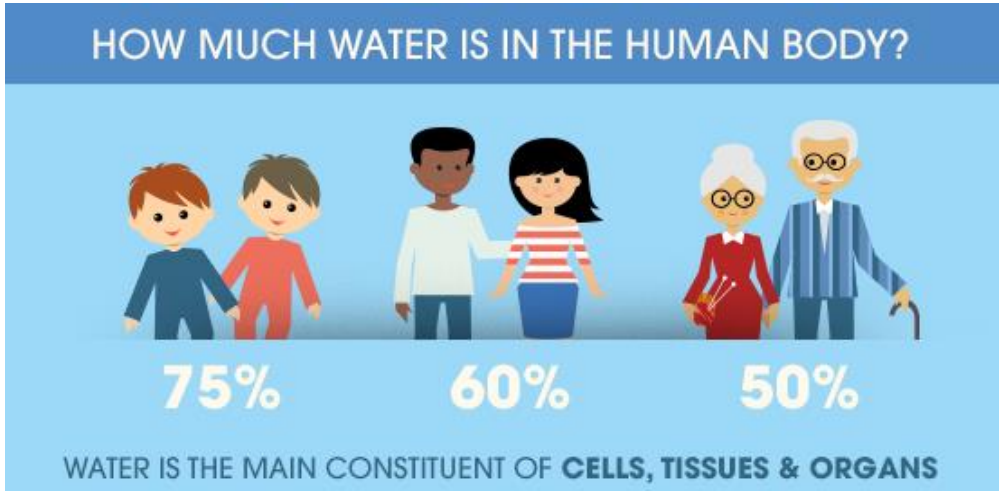
- Voiding or emptying of the urinary bladder
- **Initiation:**
  - Stretch receptors in bladder wall send signals to **sacral spinal cord**
  - Reflex impulses return via **pelvic splanchnic nerves** → bladder contracts
- **Urge to void:**
  - Urine pushed past **involuntary internal sphincter**
  - Reaches **upper urethra**
- **Control:**
  - **External urethral sphincter** is **voluntary**
  - Micturition can be **delayed consciously**

BODY FLUID COMPARTMENT

BODY FLUID COMPARTMENT

💧 **Body Water Content by Age & Sex**

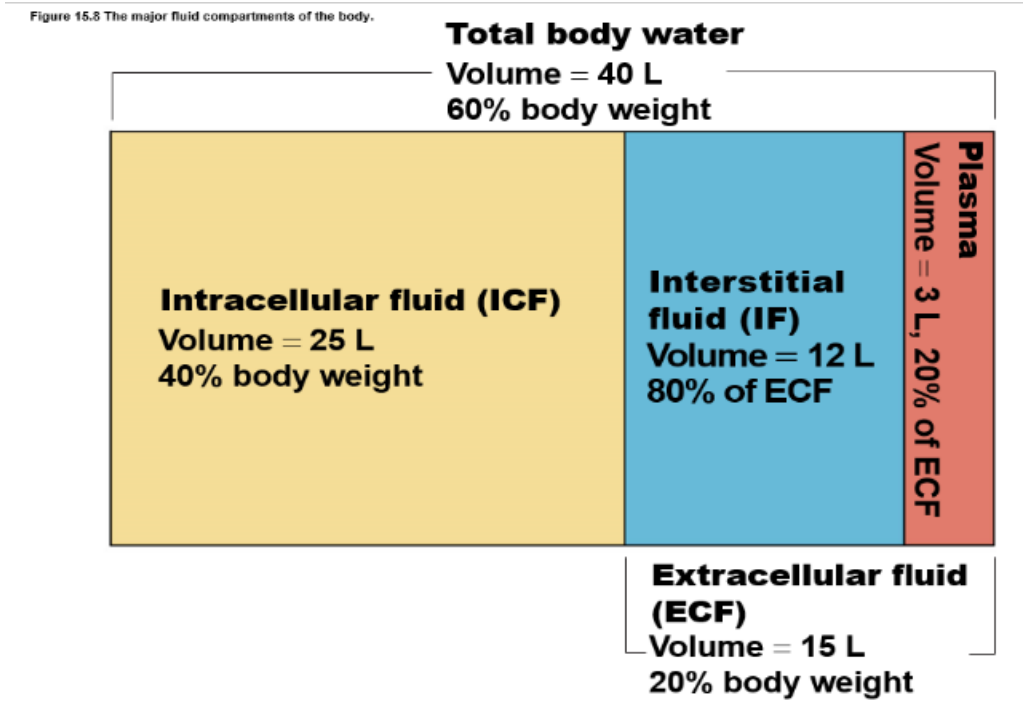
- **Babies:** ~75%
- **Young adult males:** ~60%
- **Young adult females:** ~50%
- **Elderly:** ~45–50%
- Water is essential for numerous physiological functions
- Fluid balance is tightly regulated



BODY FLUID COMPARTMENT

Body Fluid Compartments

- **Intracellular Fluid (ICF)**
  - Inside cells
  - ~⅔ of total body water
- **Extracellular Fluid (ECF)**
  - Outside cells
  - Includes:
    - **Interstitial fluid (IF)**
    - **Blood plasma**
    - **Lymph**
    - **Transcellular fluids** (e.g., cerebrospinal, synovial)
- **Plasma**
  - Subdivision of ECF
  - ~3 liters of total body water



MECHANISMS OF ACID BASE BALANCE REGULATION

MECHANISMS OF ACID BASE BALANCE REGULATION

Blood Composition Influencers

- Depends on:
  - **Diet**
  - **Cellular metabolism**
  - **Urine output**

Kidney's Roles in Blood Composition

- Excretes **nitrogenous wastes**
- Regulates **water balance**
- Maintains **electrolyte levels**
- Controls **blood pH**

MECHANISMS OF ACID BASE BALANCE REGULATION

Water & Electrolyte Link

- **Electrolytes** = charged ions in solution (e.g., **Na^+**, **K^+**, **Ca^{2+}**)
- Essential for **nerve impulse transmission** and **muscle contraction**

Water Intake vs Output

- Intake **must equal** output
- Sources of intake:
  - **Food & fluids**
  - **Metabolic water** (~10%)

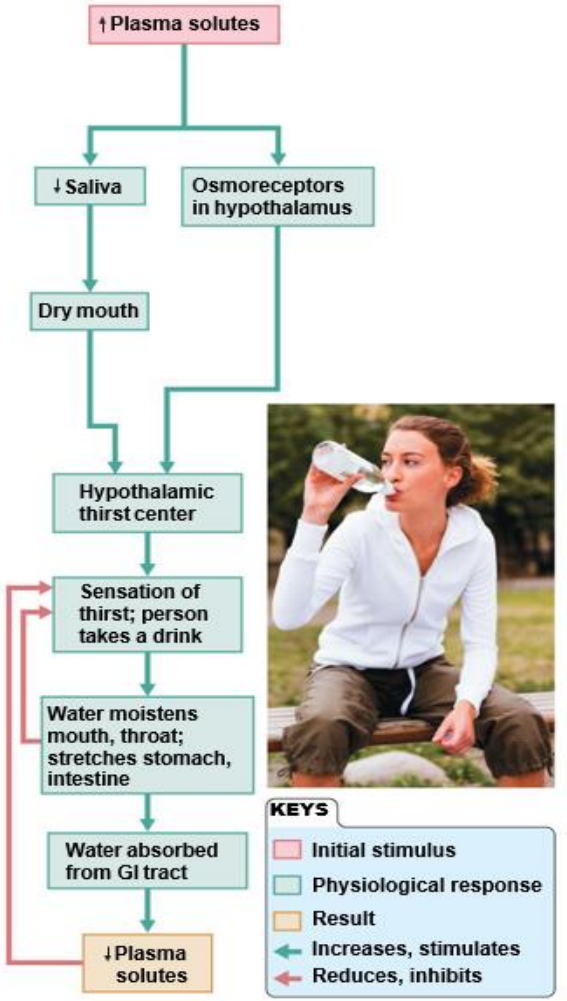
THIRST MECHANISM AND WATER OUTPUT

Thirst Mechanism

- Primary driving force for water intake
- **Osmoreceptors** in hypothalamus detect plasma solute changes
- Signals thirst center → triggers urge to drink
- **Dry mouth** from reduced saliva also triggers thirst

Sources of Water Output

- **Lungs** – insensible water loss through breathing
- **Perspiration** – varies with activity and temperature
- **Feces** – small, consistent loss
- **Urine** – major regulated route for fluid loss



HORMONAL REGULATION OF WATER & ELECTROLYTE REABSORPTION

HORMONAL REGULATION OF WATER & ELECTROLYTE REABSORPTION

Hormones: ADH & Aldosterone

👉 **Antidiuretic Hormone (ADH)**

- Made by hypothalamus, stored in posterior pituitary
- Prevents water loss by promoting reabsorption in collecting ducts
- ↓ Urine volume, ↑ blood pressure
- In high amounts, constricts arterioles (vasopressin effect)

👉 **Aldosterone**

- Regulates sodium, potassium, and water balance
- Na^+ reabsorption → Cl^- follows → K^+ secreted
- Water follows sodium → ↑ blood volume

💧 Small shifts in electrolytes cause water movement between fluid compartments

RENIN-ANGIOTENSIN MECHANISM

RENIN-ANGIOTENSIN MECHANISM

Renin-Angiotensin Mechanism

- **Main trigger for aldosterone release**
- Initiated by the **Juxtaglomerular (JG) apparatus**
- **Renin** is released in response to **low blood pressure**
- Renin → activates **angiotensin I → angiotensin II**
- **Angiotensin II:**
  - Causes **vasoconstriction**
  - Stimulates **aldosterone release**
- End result: ↑ **Blood pressure & blood volume**

ACID-BASE BALANCE & BLOOD PH REGULATION

ACID-BASE BALANCE & BLOOD PH REGULATION

🧪 Blood pH Homeostasis

- Normal range: **7.35–7.45**
- **Alkalosis:** pH > 7.45
- **Acidosis:** pH < 7.35
- **Physiological acidosis:** 7.0–7.35

Main Regulators:

- **Kidneys** – primary long-term control
- **Blood buffers**
- **Respiratory system**

ACID-BASE BALANCE & BLOOD PH REGULATION

Buffer Systems (act quickly)

- **Bicarbonate, Phosphate, Protein** buffers
- Neutralize excess H+ or OH– to stabilize pH

ACID-BASE BALANCE & BLOOD PH REGULATION

Respiratory Control

- ↑ Respiration = ↓ CO_2 = ↑ pH
- ↓ Respiration = ↑ CO_2 = ↓ pH

ACID-BASE BALANCE & BLOOD PH REGULATION

Renal Control

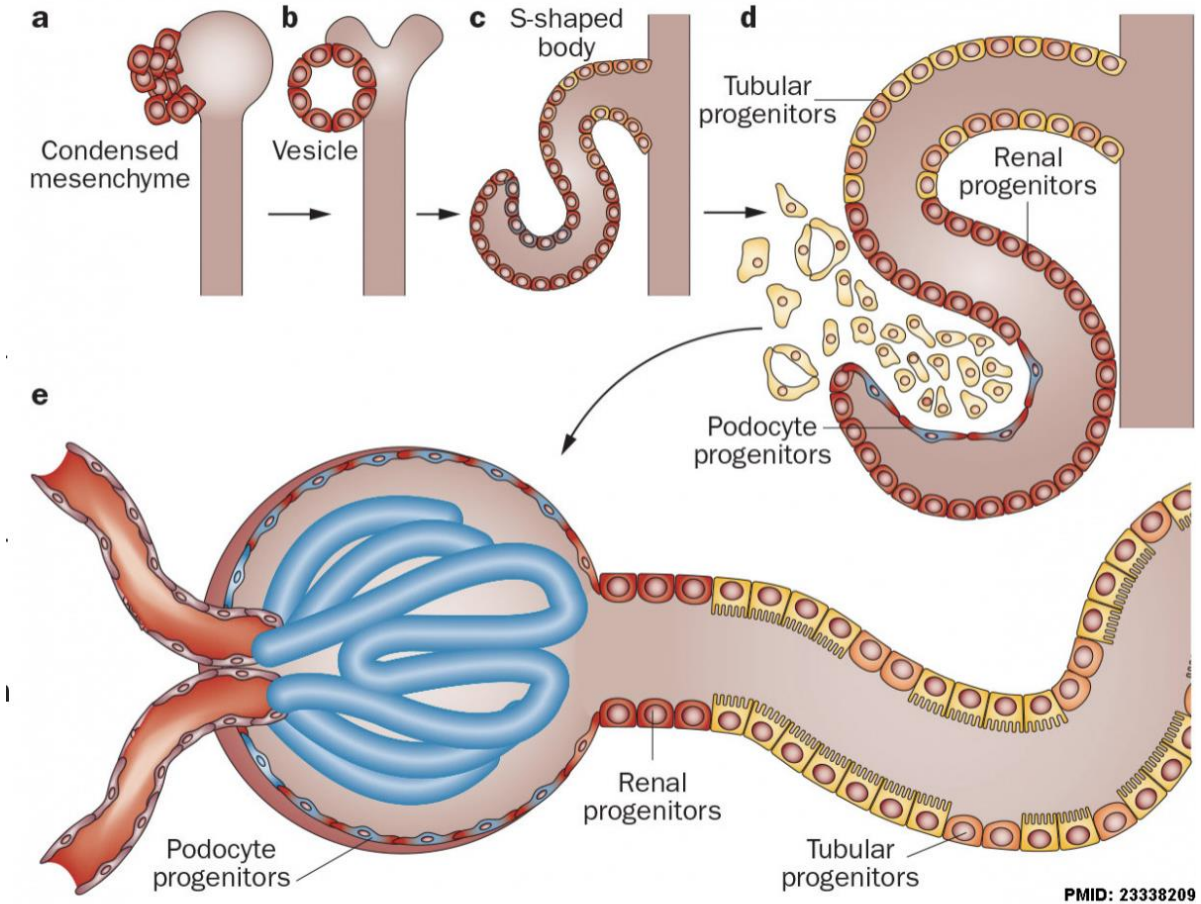
- If pH **rises:**
  - Excrete **bicarbonate,** retain **H^+**
- If pH **drops:**
  - Reabsorb **bicarbonate,** excrete **H^+**

DEVELOPMENTAL ASPECTS OF THE URINARY SYSTEM

DEVELOPMENTAL ASPECTS OF THE URINARY SYSTEM

Fetal Development of the Kidneys

- Kidney development begins early in embryonic life
- Functional urine production by third month of fetal life

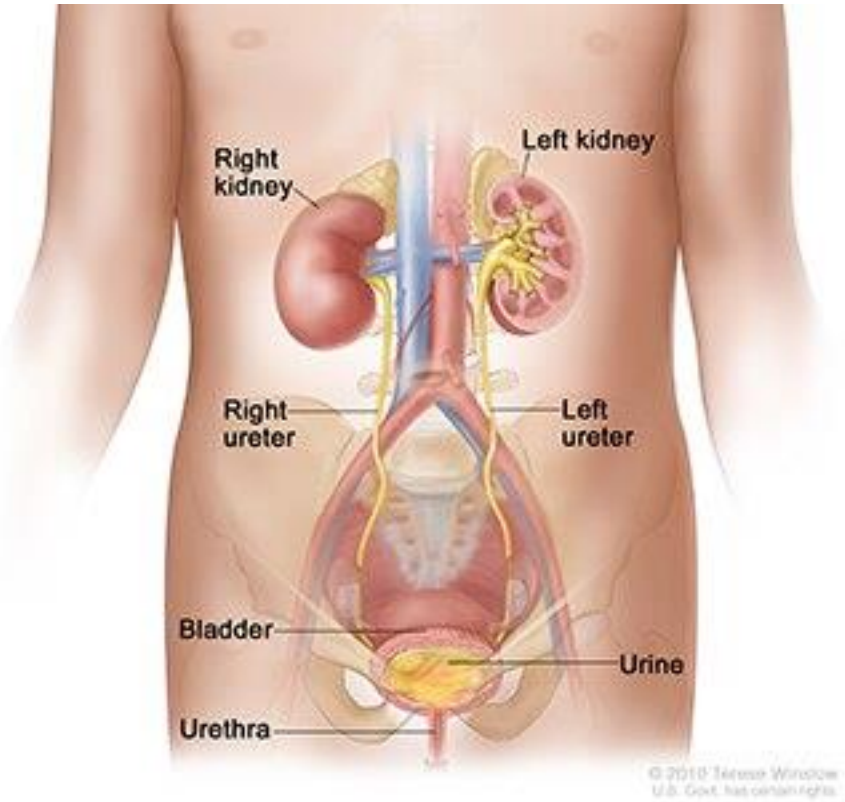


Nephron development[13]

DEVELOPMENTAL ASPECTS OF THE URINARY SYSTEM

Pediatric Concerns

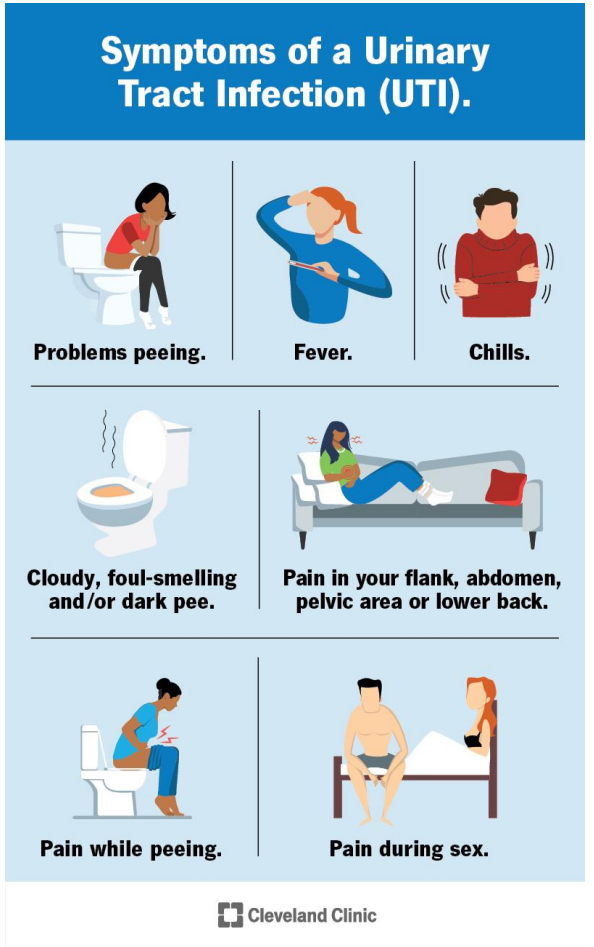
- Urinary system problems in children include:
  - Infections from fecal bacteria
  - STIs in adolescents
  - Streptococcus-related issues
- Voluntary sphincter control begins ~18 months
- Nighttime bladder control may take until age 4



DEVELOPMENTAL ASPECTS OF THE URINARY SYSTEM

Common Urinary Disorders Before Adulthood

- UTIs most common urinary issue before old age
- E. coli is responsible for ~80% of UTIs
- Symptoms: dysuria, urgency, and sometimes fever



DEVELOPMENTAL ASPECTS OF THE URINARY SYSTEM

Renal Failure in Younger Adults

- Rare but serious
- Kidneys cannot concentrate urine
- Requires dialysis for blood chemical balance

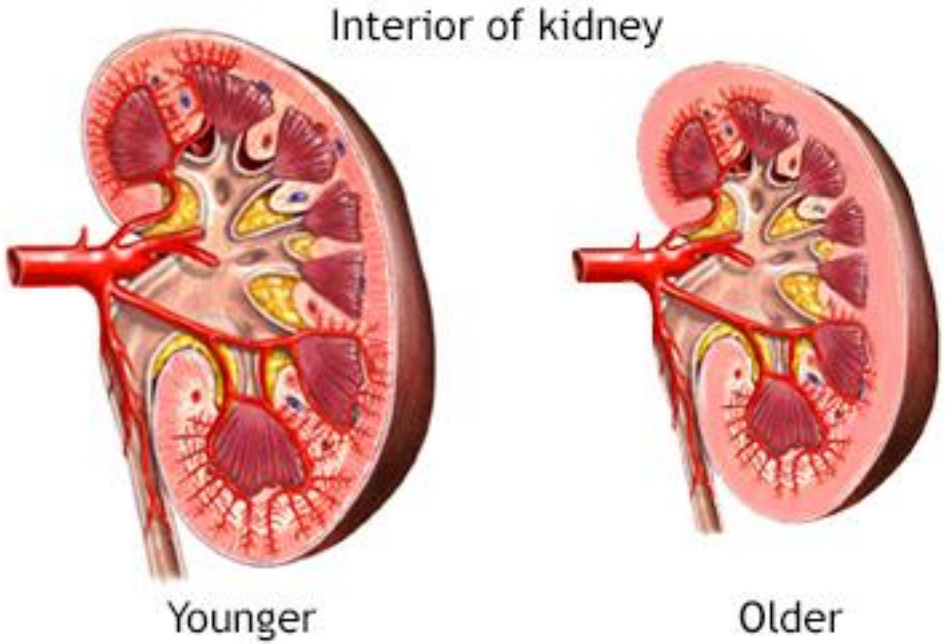


Image Source: Walsh, P. (2016, November 14). *Health insurers say dialysis clinics are scamming them.* MPR News. https://www.mprnews.org/story/2016/11/14/health-insurers-dialysis-clinics-scam

DEVELOPMENTAL ASPECTS OF THE URINARY SYSTEM

Aging and the Urinary System

- Filtration rate decreases with age
- Tubules lose efficiency
- Results in urgency, frequency, incontinence

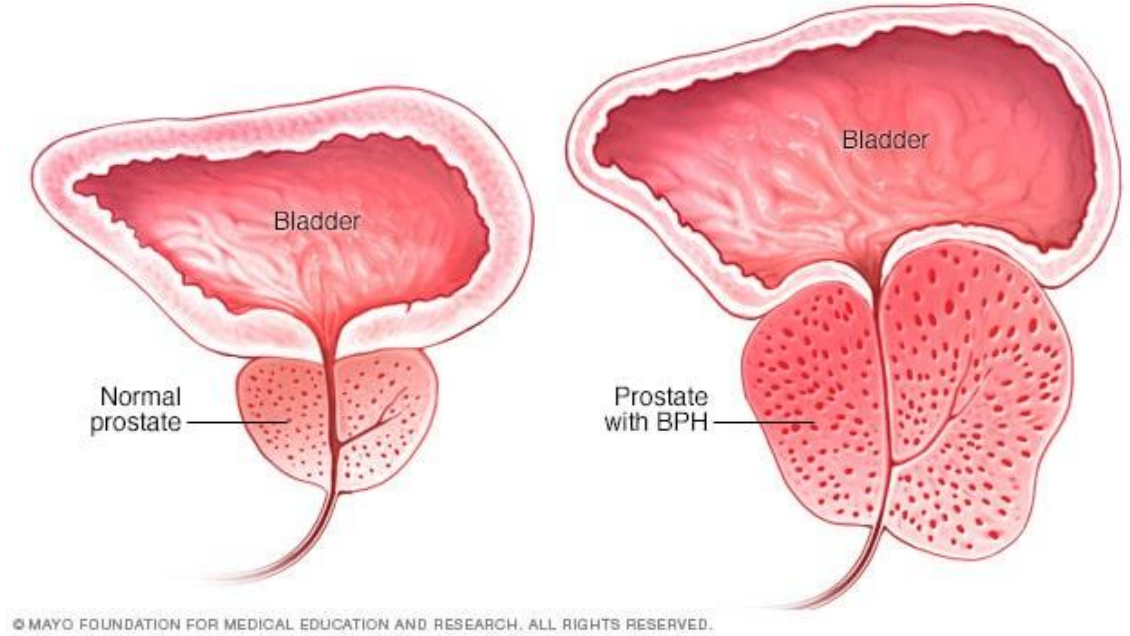


A.D.A.M.

DEVELOPMENTAL ASPECTS OF THE URINARY SYSTEM

Male-Specific Aging Concerns

- Urinary retention is common
- Often due to prostate gland enlargement



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DEVELOPMENTAL ASPECTS OF THE URINARY SYSTEM

Summary of Urinary Aging Issues

- **Urgency:** Sudden need to urinate
- **Frequency:** Frequent urination, small amounts
- **Nocturia:** Nighttime urination
- **Incontinence:** Loss of bladder control
- **Urinary Retention:** Incomplete emptying, especially in males

DEVELOPMENTAL ASPECTS OF THE URINARY SYSTEM

Conclusion

- Urinary development begins early in life
- Control and function evolve through childhood
- Aging brings functional decline and common issues
- Nurses play a critical role in detection and management